Whole numbers and estimation

Day 01 • 2026-10-29 • 15 questions

Learner PDF page 1 • W01-L01-Q01

01 A crew needs 198 studs for one wall and 403 studs for another. Round each number to the nearest hundred, then estimate the total.

600 studs

198 rounds to 200 and 403 rounds to 400. Estimate: 200 + 400 = 600 studs. The exact total, 601, is close to it.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Learner PDF page 1 • W01-L01-Q02

02 Which operation finds the unused part of a stock length?

B. Subtract used length from stock length

An offcut is the whole stock length minus the lengths used.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Learner PDF page 1 • W01-L01-Q03

03 In 4,730, what does the digit 7 represent?

D. 700

The 7 is in the hundreds place, so its value is 700.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Learner PDF page 2 • W01-L01-Q05

04 Which unit belongs with a measured board length?

B. mm

A length uses a linear unit such as millimetres.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Learner PDF page 2 • W01-L01-Q07

05 Which calculation splits 360 nails equally among 9 boxes?

A. 360 ÷ 9

Division finds the number in each equal group.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Whole numbers and estimation

Day 01 • 2026-10-29 • 15 questions

Learner PDF page 3 • W01-L01-Q09

06 Use the Shape Canvas rectangle. Add its 1,200 mm length and 800 mm width.

2000 mm

1,200 + 800 = 2,000 mm; this is the sum of two labelled lengths.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Learner PDF page 3 • W01-L01-Q10

07 A delivery contains 275 screws in one box and 438 in another. How many screws arrived?

713 screws

275 + 438 = 713 screws.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Learner PDF page 3 • W01-L01-Q11

08 A 3,600 mm board has 1,450 mm used. How much remains?

2150 mm

3,600 − 1,450 = 2,150 mm.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Learner PDF page 4 • W01-L01-Q13

09 Share 432 screws equally among 12 containers. How many screws go in each?

36 screws

432 ÷ 12 = 36 screws per container.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Learner PDF page 4 • W01-L01-Q15

10 Estimate 49 × 21 by rounding both numbers to the nearest ten.

1000

49 rounds to 50 and 21 rounds to 20; 50 × 20 = 1,000.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Whole numbers and estimation

Day 01 • 2026-10-29 • 15 questions

Learner PDF page 4 • W01-L01-Q17

11 The Shape Canvas rectangle has sides labelled 1,200 mm and 800 mm. How much longer is the longer label?

400 mm

1,200 − 800 = 400 mm.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Learner PDF page 5 • W01-L01-Q18

12 A crew orders 7 packs of 48 brackets. How many brackets are ordered?

336 brackets

7 × 48 = 336 brackets.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Learner PDF page 5 • W01-L01-Q19

13 A learner says 5 × 198 is about 100. Which estimate exposes the error?

C. 5 × 200 = 1,000

The nearby product is 1,000, so an answer near 100 is too small.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Learner PDF page 5 • W01-L01-Q21

14 Each pallet holds 125 blocks. How many blocks are on four full pallets?

500 blocks

4 × 125 = 500 blocks.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Learner PDF page 5 • W01-L01-Q23

15 A bag contains 500 nails. After 176 and then 89 are used, how many remain?

235 nails

176 + 89 = 265 nails used; 500 − 265 = 235.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Calculator basics and checking

Day 01 • 2026-10-29 • 15 questions

Learner PDF page 1 • W01-L02-Q01

01 Estimate 48 × 21 by rounding each number to the nearest ten first.

1,000

48 rounds to 50 and 21 rounds to 20. Estimate: 50 × 20 = 1,000. The exact answer, 1,008, is close.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Learner PDF page 1 • W01-L02-Q02

02 Which check reverses 1,680 ÷ 24 = 70?

B. 70 × 24 = 1,680

Multiplication reverses division and reconstructs the original dividend.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Learner PDF page 1 • W01-L02-Q03

03 What does 17^2 mean?

B. 17 × 17

Squaring multiplies a number by itself.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Learner PDF page 2 • W01-L02-Q05

04 A calculator shows 36 × 25 = 900. Check it with the inverse operation: what is 900 ÷ 25?

36

Division undoes multiplication: 900 ÷ 25 = 36. That matches the first number, so 900 is correct.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Learner PDF page 2 • W01-L02-Q07

05 A negative entry is needed. Which key purpose should you look for?

A. Change sign

Change sign changes the sign of the current number; subtraction is an operation between quantities.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Calculator basics and checking

Day 01 • 2026-10-29 • 15 questions

Learner PDF page 3 • W01-L02-Q09

06 Check the difference between the two labelled Shape Canvas dimensions: 1,200 mm minus 800 mm.

400 mm

1,200 − 800 = 400 mm; adding 400 to 800 checks it.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Learner PDF page 3 • W01-L02-Q10

07 Calculate 27 × 16.

432

27 × 16 = 432; 27 × (10 + 6) checks the result.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Learner PDF page 3 • W01-L02-Q11

08 Calculate 2,160 ÷ 18.

120

2,160 ÷ 18 = 120; 120 × 18 = 2,160.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Learner PDF page 4 • W01-L02-Q13

09 Calculate the positive square root of 784.

28

28 × 28 = 784, so √784 = 28.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Learner PDF page 4 • W01-L02-Q15

10 Calculate 48 × 25.

1200

48 × 25 = 1,200; four lots of 25 make 100, and 48 is twelve groups of four.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Calculator basics and checking

Day 01 • 2026-10-29 • 15 questions

Learner PDF page 4 • W01-L02-Q17

11 Use the Shape Canvas square, labelled 900 mm per side. Four separate pieces each match that labelled length. Find their total length.

3600 mm

Four 900 mm pieces total 4 × 900 = 3,600 mm.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Learner PDF page 5 • W01-L02-Q18

12 A learner enters 36 × 14 and gets 5,040. Which written check is strongest?

B. 36 × 10 + 36 × 4 = 504

Splitting 14 into 10 and 4 gives 360 + 144 = 504, exposing the extra zero.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Learner PDF page 5 • W01-L02-Q19

13 An invoice lists 15 boxes with 64 items each. Find the total item count.

960 items

15 × 64 = 960 items; 10 boxes give 640 and 5 give 320.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Learner PDF page 5 • W01-L02-Q21

14 A cut-list total is 6,840 mm for 12 equal pieces. Find the length of one piece.

570 mm

6,840 ÷ 12 = 570 mm per piece; 570 × 12 = 6,840.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Learner PDF page 5 • W01-L02-Q23

15 Calculate 125 × 32, then report the exact whole-number answer.

4000

125 × 32 = 4,000; 125 × 8 = 1,000 and 32 contains four groups of eight.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Quantities, averages and word problems

Day 02 • 2026-10-30 • 15 questions

Learner PDF page 1 • W01-L03-Q01

01 When checking an average in Numeracy Hub, which calculation should you enter?

B. Total of the values ÷ number of values

The mean shares the total equally among the observations.

Textbook Chapter 1 pp.17–23

Learner PDF page 2 • W01-L03-Q02

02 A problem asks for the combined lengths of different pieces. Which operation is needed first?

C. Addition

Addition combines the listed lengths into a total.

Textbook Chapter 1 pp.17–23

Learner PDF page 2 • W01-L03-Q03

03 What information is required to find the mean of five measured lengths?

B. Their sum and the count 5

Mean = sum of measurements ÷ number of measurements.

Textbook Chapter 1 pp.17–23

Learner PDF page 2 • W01-L03-Q05

04 Why should all lengths use the same unit before being added?

C. The quantities must be comparable

Adding unlike unit labels without conversion can produce a meaningless total.

Textbook Chapter 1 pp.17–23

Learner PDF page 3 • W01-L03-Q07

05 What is a useful first step in a word problem?

D. Identify the required quantity and given data

Naming the unknown and the data helps select the correct operation.

Textbook Chapter 1 pp.17–23

Quantities, averages and word problems

Day 02 • 2026-10-30 • 15 questions

Learner PDF page 3 • W01-L03-Q09

06 Use the labelled Shape Canvas lengths 1,200 mm and 800 mm. What is their mean?

1000 mm

Their total is 2,000 mm; dividing by two gives 1,000 mm.

Textbook Chapter 1 pp.17–23

Learner PDF page 4 • W01-L03-Q10

07 Three batches contain 48, 52 and 56 pieces. Find the mean batch count.

52 pieces

48 + 52 + 56 = 156; 156 ÷ 3 = 52.

Textbook Chapter 1 pp.17–23

Learner PDF page 4 • W01-L03-Q11

08 Five equal kits use 135 screws altogether. How many screws are in each kit?

27 screws

135 ÷ 5 = 27 screws per kit.

Textbook Chapter 1 pp.17–23

Learner PDF page 4 • W01-L03-Q13

09 A crew fits 18, 23 and 21 brackets on three tasks. Find the total.

62 brackets

18 + 23 + 21 = 62 brackets.

Textbook Chapter 1 pp.17–23

Learner PDF page 4 • W01-L03-Q15

10 A 2,400 mm rail is divided into six equal lengths. Ignore kerf. Find one length.

400 mm

2,400 ÷ 6 = 400 mm per length.

Textbook Chapter 1 pp.17–23

Quantities, averages and word problems

Day 02 • 2026-10-30 • 15 questions

Learner PDF page 5 • W01-L03-Q17

11 The Shape Canvas square shows a 900 mm side. Five strips each match that side length. Find their total length.

4500 mm

Five equal lengths give 5 × 900 = 4,500 mm.

Textbook Chapter 1 pp.17–23

Learner PDF page 5 • W01-L03-Q18

12 A learner finds the mean of 40, 50 and 60 as 150. What step is missing?

B. Divide the sum by 3

150 is the sum; the mean is 150 ÷ 3 = 50.

Textbook Chapter 1 pp.17–23

Learner PDF page 6 • W01-L03-Q19

13 A delivery has 8 boxes of 24 hinges. After 57 are used, how many remain?

135 hinges

The delivery contains 192 hinges; 192 − 57 = 135 remain.

Textbook Chapter 1 pp.17–23

Learner PDF page 6 • W01-L03-Q21

14 The measured lengths are 498, 501 and 500 mm. Which mean is plausible?

B. About 500 mm

The mean must be near the three measurements and between 498 and 501 mm.

Textbook Chapter 1 pp.17–23

Learner PDF page 6 • W01-L03-Q23

15 A job needs 14 braces for each of 9 frames. Five spare braces are also required. Find the total.

131 braces

9 × 14 = 126 for the frames; 126 + 5 = 131 braces.

Textbook Chapter 1 pp.17–23

Cut-list practice and rounding

Day 02 • 2026-10-30 • 15 questions

Learner PDF page 1 • W01-L04-Q01

01 When checking a cut list in Numeracy Hub, what must the entered total include?

D. Every required piece with its quantity

A cut list combines each length multiplied by the number required.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Learner PDF page 2 • W01-L04-Q02

02 A stock count is 6.2 boards and only whole boards can be purchased. What should be ordered?

C. 7 boards

A purchase count must round upward when a fraction of an extra board is needed.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Learner PDF page 2 • W01-L04-Q03

03 What does rounding 2,463 mm to the nearest 10 mm give?

A. 2,460 mm

The ones digit is 3, so round down to 2,460 mm.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Learner PDF page 2 • W01-L04-Q05

04 Which operation checks that a cut plan fits the stock?

A. Compare total required length with stock length

The required lengths must fit within the available stock.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Learner PDF page 3 • W01-L04-Q07

05 Rounding to the nearest unit and rounding up for purchase are…

A. Different decisions based on the task

Measurement rounding finds a nearby value; a purchase count must provide enough material.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Cut-list practice and rounding

Day 02 • 2026-10-30 • 15 questions

Learner PDF page 3 • W01-L04-Q09

06 Use the Shape Canvas labels 1,200 mm and 800 mm. A cut list needs two pieces of each labelled length. Ignore kerf. Find the total.

4000 mm

2 × 1,200 + 2 × 800 = 4,000 mm.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Learner PDF page 4 • W01-L04-Q10

07 Three 640 mm pieces and two 850 mm pieces are required. Ignore kerf. Find the total length.

3620 mm

1,920 + 1,700 = 3,620 mm.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Learner PDF page 4 • W01-L04-Q11

08 Round 7,286 mm to the nearest 100 mm.

7300 mm

The tens digit is 8, so 7,286 rounds up to 7,300 mm.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Learner PDF page 4 • W01-L04-Q13

09 A task needs 83 brackets sold in packs of 12. How many full packs must be purchased?

7 packs

Six packs provide 72; seven provide 84, enough for 83.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Learner PDF page 4 • W01-L04-Q15

10 Round 12,643 to the nearest thousand.

13000

The hundreds digit is 6, so the nearest thousand is 13,000.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Cut-list practice and rounding

Day 02 • 2026-10-30 • 15 questions

Learner PDF page 5 • W01-L04-Q17

11 The Shape Canvas square has a 900 mm side. A 5,000 mm stock length supplies five pieces of that length. Ignore kerf. What remains?

500 mm

Five pieces use 4,500 mm; 500 mm remains.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Learner PDF page 5 • W01-L04-Q18

12 A cut plan needs 4,820 mm but the stock is 4,800 mm. Which conclusion is correct?

C. It is 20 mm too long before any cut loss

4,820 − 4,800 = 20 mm; rounding cannot make the physical stock longer.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Learner PDF page 6 • W01-L04-Q19

13 A crew needs 38 full pieces. Each stock length supplies 6 pieces. How many stock lengths are needed?

7 stock lengths

Six lengths give 36 pieces; seven give 42, so seven are needed.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Learner PDF page 6 • W01-L04-Q21

14 Two 1,150 mm pieces and three 620 mm pieces are planned. Stock is 4,800 mm and total cut loss is 15 mm. Find the offcut.

625 mm

Pieces use 4,160 mm; adding 15 mm loss gives 4,175 mm; 625 mm remains.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

Learner PDF page 6 • W01-L04-Q23

15 A 6,000 mm stock length loses 18 mm to the stated cuts. The pieces total 5,430 mm. Find the offcut.

552 mm

6,000 − 18 − 5,430 = 552 mm.

Textbook sections 1.4–1.8 pp.14–23; sections 2.11 p.59

BEDMAS: choosing the correct order

Day 03 • 2026-11-02 • 15 questions

Learner PDF page 1 • W02-L01-Q01

01 Calculate 20 − 4 × 3.

8

BEDMAS: multiply before you subtract. 4 × 3 = 12, then 20 − 12 = 8.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

Learner PDF page 1 • W02-L01-Q02

02 In 8 + 3 × 4, which operation comes first?

D. 3 × 4

Multiplication precedes addition when no brackets change the order.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

Learner PDF page 1 • W02-L01-Q03

03 In (8 + 3) × 4, what comes first?

D. 8 + 3

The calculation inside brackets is completed first.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

Learner PDF page 2 • W02-L01-Q05

04 How are addition and subtraction handled at the same level?

C. From left to right

Addition and subtraction have equal priority.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

Learner PDF page 2 • W02-L01-Q07

05 Which expression adds 6 to the product of 4 and 3?

B. 6 + 4 × 3

The product 4 × 3 is formed before adding 6.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

BEDMAS: choosing the correct order

Day 03 • 2026-11-02 • 15 questions

Learner PDF page 3 • W02-L01-Q09

06 Use the Shape Canvas lengths 1,200 mm and 800 mm. Calculate 1,200 + 2 × 800.

2800 mm

Multiply first: 2 × 800 = 1,600; then add 1,200 to get 2,800 mm.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

Learner PDF page 3 • W02-L01-Q10

07 Calculate 7 + 5 × 6.

37

5 × 6 = 30; 7 + 30 = 37.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

Learner PDF page 3 • W02-L01-Q11

08 Calculate (7 + 5) × 6.

72

Brackets give 12; 12 × 6 = 72.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

Learner PDF page 4 • W02-L01-Q13

09 Calculate 20 − 8 + 3.

15

Work left to right: 20 − 8 = 12, then 12 + 3 = 15.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

Learner PDF page 4 • W02-L01-Q15

10 Calculate 48 ÷ (4 + 2).

8

The bracket is 6; 48 ÷ 6 = 8.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

BEDMAS: choosing the correct order

Day 03 • 2026-11-02 • 15 questions

Learner PDF page 4 • W02-L01-Q17

11 The Shape Canvas square has a 900 mm side. Two pieces match that length and a third is 300 mm. Find 2 × 900 + 300.

2100 mm

Multiply first: 1,800 + 300 = 2,100 mm.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

Learner PDF page 5 • W02-L01-Q18

12 A learner obtains 48 for 6 + 2 × 6. What is the error?

A. Adding before multiplying

The correct order gives 2 × 6 = 12, then 6 + 12 = 18.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

Learner PDF page 5 • W02-L01-Q19

13 A job uses 4 packs of 25 screws plus 18 loose screws. Find the total.

118 screws

4 × 25 = 100; adding 18 gives 118 screws.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

Learner PDF page 5 • W02-L01-Q21

14 A stock length is 2,400 mm. Three pieces of 650 mm are used. Ignore kerf. Calculate 2,400 − 3 × 650.

450 mm

The pieces use 1,950 mm; 2,400 − 1,950 = 450 mm.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

Learner PDF page 5 • W02-L01-Q23

15 Calculate 5 × (14 + 6) − 12.

88

The brackets give 20; 5 × 20 = 100; subtracting 12 gives 88.

Textbook Chapter 3 pp.68–75; Chapter 2 pp.51–53

BEDMAS in cut lists and frames

Day 03 • 2026-11-02 • 15 questions

Learner PDF page 1 • W02-L02-Q01

01 When using Numeracy Hub to check a frame expression, what should the written expression preserve?

B. The quantities, brackets and operations from the plan

The tool can only check the calculation actually entered.

Textbook Chapter 3 pp.68–75

Learner PDF page 2 • W02-L02-Q02

02 A plan needs two pieces of length L and two of width W. Which expression totals the pieces?

C. 2 × L + 2 × W

Each of the two lengths and two widths must be counted.

Textbook Chapter 3 pp.68–75

Learner PDF page 2 • W02-L02-Q03

03 Which expression means three identical sets, each containing a 600 mm and a 400 mm piece?

B. 3 × (600 + 400)

The bracket is one complete set; multiplication counts three sets.

Textbook Chapter 3 pp.68–75

Learner PDF page 2 • W02-L02-Q05

04 What does 4 × (350 + 250) count?

B. Four complete pairs of pieces

The bracket groups both pieces before multiplying the set count.

Textbook Chapter 3 pp.68–75

Learner PDF page 3 • W02-L02-Q07

05 How should a stated total kerf allowance K enter an offcut expression?

D. Stock − piece total − K

Kerf consumes stock and must also be subtracted.

Textbook Chapter 3 pp.68–75

BEDMAS in cut lists and frames

Day 03 • 2026-11-02 • 15 questions

Learner PDF page 3 • W02-L02-Q09

06 Using the Shape Canvas rectangle labels, total two 1,200 mm pieces and two 800 mm pieces. Ignore joints and kerf.

4000 mm

One pair is 2,000 mm; two pairs total 4,000 mm.

Textbook Chapter 3 pp.68–75

Learner PDF page 4 • W02-L02-Q10

07 Three sets each require a 700 mm piece and a 450 mm piece. Find the total length.

3450 mm

One set is 1,150 mm; three sets use 3,450 mm.

Textbook Chapter 3 pp.68–75

Learner PDF page 4 • W02-L02-Q11

08 A 4,800 mm stock length supplies five 760 mm pieces. Ignore kerf. Find the offcut.

1000 mm

Five pieces use 3,800 mm, leaving 1,000 mm.

Textbook Chapter 3 pp.68–75

Learner PDF page 4 • W02-L02-Q13

09 A plan needs two 1,350 mm pieces and two 650 mm pieces. Find the total.

4000 mm

2 × 2,000 = 4,000 mm.

Textbook Chapter 3 pp.68–75

Learner PDF page 4 • W02-L02-Q15

10 Four identical kits contain 12 large screws and 8 small screws each. Find the total screw count.

80 screws

Each kit contains 20 screws; four kits contain 80.

Textbook Chapter 3 pp.68–75

BEDMAS in cut lists and frames

Day 03 • 2026-11-02 • 15 questions

Learner PDF page 5 • W02-L02-Q17

11 The Shape Canvas square shows 900 mm. A kit has three pieces of this length and two 300 mm pieces. Find the total.

3300 mm

2,700 + 600 = 3,300 mm.

Textbook Chapter 3 pp.68–75

Learner PDF page 5 • W02-L02-Q18

12 A learner enters 4 × 600 + 400 for four complete 600 mm/400 mm pairs. What correction is needed?

D. Use 4 × (600 + 400)

Both pieces belong to each pair, so the whole pair is multiplied by 4.

Textbook Chapter 3 pp.68–75

Learner PDF page 6 • W02-L02-Q19

13 A 6,000 mm stock length supplies two 1,250 mm pieces and three 900 mm pieces. Total cut loss is 20 mm. Find the offcut.

780 mm

Pieces use 5,200 mm; including loss uses 5,220 mm, leaving 780 mm.

Textbook Chapter 3 pp.68–75

Learner PDF page 6 • W02-L02-Q21

14 A box holds 200 brackets. Six kits each need 14 brackets. Find the remaining count.

116 brackets

Six kits use 84 brackets; 200 − 84 = 116 remain.

Textbook Chapter 3 pp.68–75

Learner PDF page 6 • W02-L02-Q23

15 Five frames each need two 420 mm pieces and two 280 mm pieces. Find the total cut length.

7000 mm

One frame needs 1,400 mm; five need 7,000 mm.

Textbook Chapter 3 pp.68–75

Brackets, squares and square roots

Day 04 • 2026-11-03 • 15 questions

Learner PDF page 1 • W02-L03-Q01

01 Calculate $3 + 4^2$.

19

The exponent acts only on the 4. $4^2 = 16$, then $3 + 16 = 19$.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Learner PDF page 1 • W02-L03-Q02

02 What is the first operation in $(6 + 4)^2$?

D. Add 6 and 4

Brackets are evaluated before squaring the bracket result.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Learner PDF page 2 • W02-L03-Q03

03 Which statement correctly interprets $\sqrt{81}$?

A. 9, because 9 × 9 = 81

The positive square root is the number whose square is 81.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Learner PDF page 2 • W02-L03-Q05

04 Which action reverses squaring a positive value?

B. Take the positive square root

Square and square root undo each other for positive values.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Learner PDF page 2 • W02-L03-Q07

05 Which expression squares the entire difference between 15 and 9?

C. $(15 - 9)^2$

Brackets make the difference the base of the square.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Brackets, squares and square roots

Day 04 • 2026-11-03 • 15 questions

Learner PDF page 3 • W02-L03-Q09

06 Read the Shape Canvas rectangle labels. The longer label is 1,200 mm and the shorter is 800 mm. Find their difference before applying any other operation.

400 mm

The bracket-like grouped difference is 1,200 − 800 = 400 mm.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Learner PDF page 3 • W02-L03-Q10

07 Calculate $(9 - 4)^2$.

25

The bracket is 5; $5^2 = 25$.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Learner PDF page 3 • W02-L03-Q11

08 Calculate $\sqrt{(49 + 32)}$.

9

The radicand is 81; $\sqrt{81} = 9$.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Learner PDF page 4 • W02-L03-Q13

09 Calculate $(6 + 2)^2$.

64

The bracket is 8; $8^2 = 64$.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Learner PDF page 4 • W02-L03-Q15

10 Calculate $100 - \sqrt{144}$.

88

$\sqrt{144} = 12$; $100 - 12 = 88$.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Brackets, squares and square roots

Day 04 • 2026-11-03 • 15 questions

Learner PDF page 4 • W02-L03-Q17

11 The Shape Canvas square has a 900 mm side label. Find the positive square root of the number 900 (no length formula is required).

30

30 × 30 = 900, so the positive root of the number is 30.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Learner PDF page 5 • W02-L03-Q18

12 A learner writes $(8 - 3)^2 = 8 - 9$. Which correction is right?

A. Subtract first, then square: $5^2 = 25$

The brackets make 8 − 3 the complete base to be squared.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Learner PDF page 5 • W02-L03-Q19

13 Calculate √(16 + 9) + 7.

12

16 + 9 = 25; √25 = 5; 5 + 7 = 12.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Learner PDF page 5 • W02-L03-Q21

14 Calculate $2^2 \times (11 - 3)$.

32

$2^2 = 4$ and the bracket is 8; 4 × 8 = 32.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Learner PDF page 5 • W02-L03-Q23

15 Calculate $(18 \div 3)^2 - 10$.

26

18 ÷ 3 = 6; $6^2 = 36$; 36 − 10 = 26.

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

Calculation checking workshop

Day 04 • 2026-11-03 • 15 questions

Learner PDF page 1 • W02-L04-Q01

01 Estimate 612 + 289 by rounding each number to the nearest hundred.

900

612 rounds to 600 and 289 rounds to 300. Estimate: 600 + 300 = 900. The exact sum, 901, is close.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Learner PDF page 1 • W02-L04-Q02

02 Which check best tests 84 ÷ 7 = 12?

B. 12 × 7 = 84

Multiplication reverses division.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Learner PDF page 1 • W02-L04-Q03

03 A learner gets 9 for 3 + 2 × 3. Is the result correct?

B. Yes: 2 × 3 = 6, then add 3

Multiplication is completed before addition, giving 9.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Learner PDF page 2 • W02-L04-Q05

04 Which comparison is strongest when a result seems too large?

D. Compare with an estimate and a reverse-operation check

Independent checks test both the scale and the calculation.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Learner PDF page 2 • W02-L04-Q07

05 Which habit improves a retry after an order-of-operations error?

C. Mark the first operation before calculating

Marking the next operation makes the corrected reasoning explicit.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Calculation checking workshop

Day 04 • 2026-11-03 • 15 questions

Learner PDF page 3 • W02-L04-Q09

06 Use the Shape Canvas rectangle labels to evaluate (1,200 + 800) ÷ 4.

500 mm

The bracket total is 2,000 mm; dividing by 4 gives 500 mm.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Learner PDF page 3 • W02-L04-Q10

07 Recalculate 9 + 4 × 7 correctly.

37

4 × 7 = 28; 9 + 28 = 37.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Learner PDF page 3 • W02-L04-Q11

08 Recalculate 72 ÷ 8 × 3 correctly.

27

Divide and multiply left to right: 9 × 3 = 27.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Learner PDF page 4 • W02-L04-Q13

09 Recalculate $(10 - 4)^2$ correctly.

36

The bracket is 6; $6^2 = 36$.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Learner PDF page 4 • W02-L04-Q15

10 Calculate 5 × (14 + 6) ÷ 4.

25

The bracket is 20; 5 × 20 = 100; 100 ÷ 4 = 25.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Calculation checking workshop

Day 04 • 2026-11-03 • 15 questions

Learner PDF page 4 • W02-L04-Q17

11 The Shape Canvas square shows a 900 mm side. A learner must divide three pieces of this length into six equal shares. Find 3 × 900 ÷ 6.

450 mm

Three lengths total 2,700 mm; dividing by six gives 450 mm.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Learner PDF page 5 • W02-L04-Q18

12 For 7 + 3 × 5, a learner wrote 10 × 5 = 50. What is the first incorrect step?

A. Adding 7 and 3 before multiplying

The product 3 × 5 must be found first; the correct final answer is 22.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Learner PDF page 5 • W02-L04-Q19

13 A 5,000 mm stock length supplies four 950 mm pieces and 16 mm total kerf. Find the offcut as a check.

1184 mm

Pieces use 3,800 mm; including 16 mm kerf leaves 1,184 mm.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Learner PDF page 5 • W02-L04-Q21

14 A learner retries a bracket problem with fresh values: 6 × (25 − 9) + 4. Find the result.

100

The bracket is 16; 6 × 16 = 96; adding 4 gives 100.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Learner PDF page 5 • W02-L04-Q23

15 Calculate (28 + 20) ÷ (9 − 3).

8

The numerator bracket is 48 and the denominator bracket is 6; 48 ÷ 6 = 8.

Textbook Chapter 2 p.59; Chapter 3 pp.68–75

Fractions and inch-tape markings

Day 05 • 2026-11-04 • 15 questions

Learner PDF page 1 • W02-L05-Q01

01 When using the Numeracy Hub measuring-tool picture, what determines a measurement?

D. The labelled scale and its divisions

A digital drawing is read from labels and scale divisions, not its physical screen size.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Learner PDF page 2 • W02-L05-Q02

02 In the fraction 3/8, what does the denominator 8 tell you?

A. The whole is divided into 8 equal parts

The denominator counts the equal parts in one whole.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Learner PDF page 2 • W02-L05-Q03

03 In 5/16, what does the numerator 5 tell you?

D. Five of the sixteen equal parts are counted

The numerator counts how many equal parts are included.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Learner PDF page 2 • W02-L05-Q05

04 Which is the smallest of these inch intervals?

A. 1/16 in

A larger denominator means smaller equal parts when each numerator is 1.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Learner PDF page 3 • W02-L05-Q07

05 Which fraction is equivalent to one whole?

D. 8/8

All eight of eight equal parts reconstruct the whole.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Fractions and inch-tape markings

Day 05 • 2026-11-04 • 15 questions

Learner PDF page 3 • W02-L05-Q09

06 The pictured rectangle’s longer side is 18½ in (18.5 in on a decimal label). What is the numerator of its 1/2-in fractional part?

1

The fractional part is 1/2; its numerator is 1.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Learner PDF page 4 • W02-L05-Q10

07 How many equal eighth-inch intervals fit in one inch?

8 intervals

Eight intervals of 1/8 inch make 8/8 inch, one whole inch.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Learner PDF page 4 • W02-L05-Q11

08 A mark is five sixteenth-inch intervals after zero. Give the reading in inches as a fraction or decimal.

5/16 in

Count five intervals of 1/16 inch: the reading is 5/16 inch.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Learner PDF page 4 • W02-L05-Q13

09 A mark lies 3 eighth-inch intervals beyond the 4-inch mark. Give the full reading.

4 3/8 in

Combine four whole inches with three eighths: 4 3/8 in.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Learner PDF page 4 • W02-L05-Q15

10 Which length is 12/16 in written in lowest terms?

B. 3/4 in

Divide numerator and denominator by 4: 12/16 = 3/4.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Fractions and inch-tape markings

Day 05 • 2026-11-04 • 15 questions

Learner PDF page 5 • W02-L05-Q17

11 The pictured rectangle’s shorter side is 7¼ in (7.25 in on a decimal label). How many full inches occur before its fractional part?

7 in

The mixed measurement 7 1/4 in contains seven whole inches.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Learner PDF page 5 • W02-L05-Q18

12 A learner counts tick marks including zero and calls the third tick 3/8 in. What should be counted instead?

B. The intervals from zero

Measurements count spaces from zero; counting zero as an interval gives an extra part.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Learner PDF page 6 • W02-L05-Q19

13 A tape is divided into sixteenths. A mark lies 10 intervals after zero. Give its simplest reading.

5/8 in

10/16 reduces by 2 to 5/8 in.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Learner PDF page 6 • W02-L05-Q21

14 A mark is 11 sixteenth-inch intervals beyond the 5-inch mark. Give the full reading.

5 11/16 in

The whole part is 5 in and the fractional part is 11/16 in.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Learner PDF page 6 • W02-L05-Q23

15 A tape has 16 intervals per inch. How many intervals separate zero from 3/4 in?

12 intervals

Three quarters of 16 intervals is 12; the mark is 12/16 = 3/4 in.

Textbook sections 4.1–4.4 pp.88–94; sections 4.5 pp.95–96

Tape and rule reading practice

Day 05 • 2026-11-04 • 15 questions

Learner PDF page 1 • W02-L06-Q01

01 What should you confirm before reading a rule pictured in Numeracy Hub?

B. Its unit and smallest division

A valid reading starts with the unit and the size of one scale interval.

Textbook Chapter 4 pp.88–92

Learner PDF page 2 • W02-L06-Q02

02 A rule has eight equal intervals in each inch. One interval is…

A. 1/8 in

One inch divided into eight equal intervals gives eighths.

Textbook Chapter 4 pp.88–92

Learner PDF page 2 • W02-L06-Q03

03 What is the safest way to identify a 3/4-in mark on a sixteenth-inch rule?

C. Locate the twelfth interval after zero

3/4 = 12/16, so it is the twelfth sixteenth-inch interval.

Textbook Chapter 4 pp.88–92

Learner PDF page 2 • W02-L06-Q05

04 Which method avoids a parallax reading error on a physical rule?

B. Look squarely at the scale and end mark

Viewing squarely avoids apparent shifts between the mark and the scale.

Textbook Chapter 4 pp.88–92

Learner PDF page 3 • W02-L06-Q07

05 Which two readings name the same mark?

D. 1/2 in and 8/16 in

Multiplying both parts of 1/2 by 8 gives 8/16.

Textbook Chapter 4 pp.88–92

Tape and rule reading practice

Day 05 • 2026-11-04 • 15 questions

Learner PDF page 3 • W02-L06-Q09

06 The pictured Shape Canvas longer side is 18½ in. Express its fractional part using a denominator of 16: what numerator is needed?

8

1/2 = 8/16, so the numerator is 8.

Textbook Chapter 4 pp.88–92

Learner PDF page 4 • W02-L06-Q10

07 On a sixteenth-inch rule, the seventh interval after 2 in is what full reading?

2 7/16 in

Add the fractional reading to the whole-inch mark: 2 7/16 in.

Textbook Chapter 4 pp.88–92

Learner PDF page 4 • W02-L06-Q11

08 On an eighth-inch rule, the fifth interval after 9 in is what full reading?

9 5/8 in

The whole part is 9 in and five eighths are counted after it.

Textbook Chapter 4 pp.88–92

Learner PDF page 4 • W02-L06-Q13

09 How many sixteenth-inch intervals make 1/4 in?

4 intervals

One quarter of sixteen is four; 1/4 = 4/16.

Textbook Chapter 4 pp.88–92

Learner PDF page 4 • W02-L06-Q15

10 A piece starts at the 2-inch mark and ends at the 9-inch mark. Find its length.

7 in

The length is end minus start: 9 − 2 = 7 in.

Textbook Chapter 4 pp.88–92

Tape and rule reading practice

Day 05 • 2026-11-04 • 15 questions

Learner PDF page 5 • W02-L06-Q17

11 The pictured rectangle’s shorter side is 7¼ in. On a sixteenth-inch rule, how many small intervals occur after 7 in?

4 intervals

The extra quarter inch equals four sixteenth-inch intervals.

Textbook Chapter 4 pp.88–92

Learner PDF page 5 • W02-L06-Q18

12 A learner reads 4 6/16 in. Which simplified label should be recorded?

C. 4 3/8 in

6/16 reduces to 3/8 without changing the measured length.

Textbook Chapter 4 pp.88–92

Learner PDF page 6 • W02-L06-Q19

13 A mark is three quarter-inch intervals after the 6-inch mark. Give the full reading.

6 3/4 in

Three quarter intervals make 3/4 inch beyond the 6-inch mark.

Textbook Chapter 4 pp.88–92

Learner PDF page 6 • W02-L06-Q21

14 A mark is 15 sixteenth-inch intervals beyond 10 in. Give the full reading.

10 15/16 in

The mark is one sixteenth short of 11 in: 10 15/16 in.

Textbook Chapter 4 pp.88–92

Learner PDF page 6 • W02-L06-Q23

15 A physical rule has 32 equal intervals per inch. How many intervals make 3/8 in?

12 intervals

3/8 = 12/32, so the mark is twelve intervals from zero.

Textbook Chapter 4 pp.88–92

Equivalent fractions and comparing lengths

Day 06 • 2026-11-05 • 15 questions

Learner PDF page 1 • W02-L07-Q01

01 When Numeracy Hub shows a rule divided into sixteenths, how can 1/2 in be represented on that scale?

C. 8/16 in

Half of sixteen equal intervals is eight.

Textbook Chapter 4 pp.91–94

Learner PDF page 2 • W02-L07-Q02

02 Which operation preserves a fraction’s value?

A. Multiply numerator and denominator by the same nonzero number

Scaling both parts equally keeps the same ratio of counted parts to total parts.

Textbook Chapter 4 pp.91–94

Learner PDF page 2 • W02-L07-Q03

03 Which fraction equals 3/4?

D. 12/16

Multiplying numerator and denominator of 3/4 by 4 gives 12/16.

Textbook Chapter 4 pp.91–94

Learner PDF page 2 • W02-L07-Q05

04 Which fraction is in simplest form?

B. 5/8

5 and 8 have no common whole-number factor greater than 1.

Textbook Chapter 4 pp.91–94

Learner PDF page 3 • W02-L07-Q07

05 Which is greater: 1/4 or 1/8?

C. 1/4

With equal numerators, the fraction made from fewer, larger parts is greater.

Textbook Chapter 4 pp.91–94

Equivalent fractions and comparing lengths

Day 06 • 2026-11-05 • 15 questions

Learner PDF page 3 • W02-L07-Q09

06 The pictured rectangle’s longer side is 18½ in. Write the 1/2-in fractional part with denominator 8: enter the numerator.

4

1/2 = 4/8, so four eighths represent the fractional part.

Textbook Chapter 4 pp.91–94

Learner PDF page 4 • W02-L07-Q10

07 Complete the equivalent fraction: 3/4 = ?/20. Enter the numerator.

15

The denominator is multiplied by 5, so multiply the numerator by 5: 15/20.

Textbook Chapter 4 pp.91–94

Learner PDF page 4 • W02-L07-Q11

08 Which fraction equals 18/24 in lowest terms?

C. 3/4

Divide both numerator and denominator by 6: 18/24 = 3/4.

Textbook Chapter 4 pp.91–94

Learner PDF page 4 • W02-L07-Q13

09 Which fraction equals 14/21 in lowest terms?

A. 2/3

Divide both parts by 7: 14/21 = 2/3.

Textbook Chapter 4 pp.91–94

Learner PDF page 5 • W02-L07-Q15

10 Which fraction equals 15/25 in lowest terms?

D. 3/5

Divide both parts by 5 to get 3/5.

Textbook Chapter 4 pp.91–94

Equivalent fractions and comparing lengths

Day 06 • 2026-11-05 • 15 questions

Learner PDF page 5 • W02-L07-Q17

11 The pictured rectangle’s shorter side is 7¼ in. Express its 1/4-in fractional part with denominator 16: enter the numerator.

4

1/4 = 4/16, so the numerator is 4.

Textbook Chapter 4 pp.91–94

Learner PDF page 6 • W02-L07-Q18

12 Which cut label is longer: 7/8 in or 13/16 in?

C. 7/8 in

7/8 = 14/16, which is greater than 13/16.

Textbook Chapter 4 pp.91–94

Learner PDF page 6 • W02-L07-Q19

13 Which fraction is equivalent to 6/10?

A. 3/5

Dividing both parts by 2 gives 3/5.

Textbook Chapter 4 pp.91–94

Learner PDF page 6 • W02-L07-Q21

14 Which pair marks the same position on an inch rule?

A. 5/8 in and 10/16 in

Doubling both parts of 5/8 gives 10/16.

Textbook Chapter 4 pp.91–94

Learner PDF page 7 • W02-L07-Q23

15 Which sequence is ordered from shortest to longest?

A. 1/4 in, 3/8 in, 1/2 in

Using eighths, the values are 2/8, 3/8 and 4/8.

Textbook Chapter 4 pp.91–94

Mixed numbers and fraction calculator entry

Day 06 • 2026-11-05 • 15 questions

Learner PDF page 1 • W02-L08-Q01

01 Write 2 3/4 as an improper fraction with denominator 4. Enter the numerator.

11

2 × 4 = 8 quarters in 2 wholes. 8 + 3 = 11, so 2 3/4 = 11/4.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Learner PDF page 1 • W02-L08-Q02

02 Which improper fraction equals 4 3/8?

C. 35/8

4 × 8 + 3 = 35 eighths; the denominator stays 8.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Learner PDF page 1 • W02-L08-Q03

03 Which mixed number equals 17/4?

B. 4 1/4

17 ÷ 4 gives 4 whole units and 1 remaining quarter.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Learner PDF page 2 • W02-L08-Q05

04 Which fraction is improper?

C. 13/8

Its numerator is larger than its denominator, so its value exceeds one.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Learner PDF page 2 • W02-L08-Q07

05 Which check verifies 29/6 = 4 5/6?

A. 4 × 6 + 5 = 29

Rebuilding the improper numerator checks the quotient and remainder.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Mixed numbers and fraction calculator entry

Day 06 • 2026-11-05 • 15 questions

Learner PDF page 3 • W02-L08-Q09

06 The pictured rectangle is 18½ in long. To write this as an improper fraction over 2, what numerator is needed?

37

18 × 2 + 1 = 37, so the length is 37/2 in.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Learner PDF page 3 • W02-L08-Q10

07 Convert 6 5/8 to an improper fraction over 8. Enter its numerator.

53

Six wholes contain 48 eighths; adding five gives 53/8.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Learner PDF page 3 • W02-L08-Q11

08 Convert 31/4 to a mixed number. Enter the whole-number part.

7

31 ÷ 4 gives 7 whole units and a remainder of 3, so the mixed number is 7 3/4.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Learner PDF page 4 • W02-L08-Q13

09 Convert 9 7/16 to an improper fraction over 16. Enter its numerator.

151

9 × 16 + 7 = 151, giving 151/16.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Learner PDF page 4 • W02-L08-Q15

10 Convert 3 11/12 to an improper fraction over 12. Enter its numerator.

47

3 × 12 + 11 = 47, so the fraction is 47/12.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Mixed numbers and fraction calculator entry

Day 06 • 2026-11-05 • 15 questions

Learner PDF page 4 • W02-L08-Q17

11 The pictured rectangle is 7¼ in wide. Express this as an improper fraction over 4: enter the numerator.

29

7 × 4 + 1 = 29, so the width is 29/4 in.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Learner PDF page 5 • W02-L08-Q18

12 A learner converts 2 3/4 to 5/4 by adding 2 + 3. What is the correct improper fraction?

B. 11/4

Two wholes are eight quarters; adding three more quarters gives 11/4.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Learner PDF page 5 • W02-L08-Q19

13 Which pair describes the same tape dimension?

D. 45/8 in and 5 5/8 in

5 × 8 + 5 = 45, so 5 5/8 equals 45/8.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Learner PDF page 5 • W02-L08-Q21

14 A learner enters a mixed fraction and the display gives a value smaller than its whole-number part. What should be checked first?

D. Whether the whole and fraction were entered in the correct sequence

A positive mixed number must exceed its whole part; a smaller result suggests an entry error.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Mixed numbers and fraction calculator entry

Day 06 • 2026-11-05 • 15 questions

Learner PDF page 6 • W02-L08-Q23

15 Which mixed form of 22/8 is simplest?

D. 2 3/4

22/8 = 2 6/8; simplify 6/8 to 3/4.

Textbook Chapter 4 pp.94–96; Chapter 2 p.54

Marking fractional dimensions

Day 07 • 2026-11-06 • 15 questions

Learner PDF page 1 • W02-L09-Q01

01 When transferring a dimension from Numeracy Hub or Shape Canvas to stock, what should determine the mark?

C. The stated dimension, read with the physical rule

A digital diagram carries dimension labels; a physical rule transfers those measurements.

Textbook Chapter 4 pp.88–96

Learner PDF page 2 • W02-L09-Q02

02 A mark must be 5 3/8 in from the end. Where should the fraction be counted?

A. After the 5-inch mark

Locate the whole inches first, then count the fractional part.

Textbook Chapter 4 pp.88–96

Learner PDF page 2 • W02-L09-Q03

03 Which fraction names the mark halfway between 1/4 and 1/2 in?

D. 3/8 in

In eighths, the two endpoints are 2/8 and 4/8, with 3/8 between them.

Textbook Chapter 4 pp.88–96

Learner PDF page 2 • W02-L09-Q05

04 Why should repeated marks use a consistent starting reference?

C. Changing the reference can shift every dimension

A consistent datum makes all mark locations comparable.

Textbook Chapter 4 pp.88–96

Learner PDF page 3 • W02-L09-Q07

05 A drawing says the marked dimension is schematic. What does that mean for measurement?

A. Use the labels; the drawn size need not be to scale

A schematic communicates relationships and dimensions without guaranteeing physical scale.

Textbook Chapter 4 pp.88–96

Marking fractional dimensions

Day 07 • 2026-11-06 • 15 questions

Learner PDF page 3 • W02-L09-Q09

06 The pictured rectangle’s longer side is 18½ in. On a tape divided into eighths, what numerator labels the fraction after 18 in?

4

1/2 = 4/8, so the fractional mark is four eighths beyond 18 in.

Textbook Chapter 4 pp.88–96

Learner PDF page 4 • W02-L09-Q10

07 A mark is 9 sixteenth-inch intervals beyond 6 in. Record the dimension.

6 9/16 in

The whole part is 6 inches and the fraction is 9/16 inch.

Textbook Chapter 4 pp.88–96

Learner PDF page 4 • W02-L09-Q11

08 A mark must be 4 3/4 in from the end. How many sixteenth-inch intervals follow the 4-inch mark?

12 intervals

3/4 = 12/16, so count twelve small intervals after 4 in.

Textbook Chapter 4 pp.88–96

Learner PDF page 4 • W02-L09-Q13

09 A mark is 5 eighth-inch intervals beyond 12 in. Record the full dimension.

12 5/8 in

The mark is 12 whole inches plus 5/8 inch.

Textbook Chapter 4 pp.88–96

Learner PDF page 4 • W02-L09-Q15

10 The drawing calls for 11 3/16 in. How many whole inches should be located before counting small intervals?

11 in

First locate the 11-inch mark, then count three sixteenth-inch intervals.

Textbook Chapter 4 pp.88–96

Marking fractional dimensions

Day 07 • 2026-11-06 • 15 questions

Learner PDF page 5 • W02-L09-Q17

11 The pictured rectangle’s shorter side is 7¼ in. On an eighth-inch rule, how many small intervals follow 7 in?

2 intervals

One quarter equals two eighths, so count two intervals after 7 in.

Textbook Chapter 4 pp.88–96

Learner PDF page 5 • W02-L09-Q18

12 A learner marks 2 3/16 in instead of 2 3/8 in. Which statement is correct?

D. The mark is too short because 3/16 is less than 3/8

3/8 = 6/16; the learner used only three sixteenths after 2 in.

Textbook Chapter 4 pp.88–96

Learner PDF page 6 • W02-L09-Q19

13 Which dimension is largest?

A. 5 7/8 in

The fractional parts in sixteenths are 14, 12, 11 and 10; 14/16 is largest.

Textbook Chapter 4 pp.88–96

Learner PDF page 6 • W02-L09-Q21

14 A layout image is enlarged to twice its on-screen width. What happens to its 7¼-in dimension label?

B. It still specifies 7¼ in

Changing display scale does not change the labelled physical dimension.

Textbook Chapter 4 pp.88–96

Learner PDF page 6 • W02-L09-Q23

15 Which record gives another worker the clearest marking instruction?

B. Mark 8 5/16 in from the left reference edge

The instruction identifies both a complete dimension and its reference edge.

Textbook Chapter 4 pp.88–96

Fraction foundations checkpoint

Day 07 • 2026-11-06 • 15 questions

Learner PDF page 1 • W02-L10-Q01

01 Simplify 6/8 to lowest terms.

B. 3/4

Divide top and bottom by 2, their greatest common factor: 6/8 = 3/4.

Textbook Chapter 4 pp.88–96

Learner PDF page 1 • W02-L10-Q02

02 Which fraction has the same value as 9/12?

D. 3/4

Dividing numerator and denominator by 3 gives 3/4.

Textbook Chapter 4 pp.88–96

Learner PDF page 1 • W02-L10-Q03

03 Which tape reading is a mixed number?

D. 6 5/8 in

A mixed number combines a whole number and a proper fraction.

Textbook Chapter 4 pp.88–96

Learner PDF page 2 • W02-L10-Q05

04 Which expression converts 7 3/8 to an improper numerator?

B. 7 × 8 + 3

Seven wholes contain 56 eighths; adding three gives 59 eighths.

Textbook Chapter 4 pp.88–96

Learner PDF page 2 • W02-L10-Q07

05 How much longer is 11/16 in than 5/8 in? Give a fraction in inches.

1/16 in

Rename 5/8 as 10/16. Then 11/16 − 10/16 = 1/16 in.

Textbook Chapter 4 pp.88–96

Fraction foundations checkpoint

Day 07 • 2026-11-06 • 15 questions

Learner PDF page 3 • W02-L10-Q09

06 The pictured rectangle’s longer side is 18½ in. Express it as an improper fraction over 2: enter the numerator.

37

18 × 2 + 1 = 37, so the dimension is 37/2 in.

Textbook Chapter 4 pp.88–96

Learner PDF page 3 • W02-L10-Q10

07 A tape mark lies 13 sixteenth-inch intervals beyond 7 in. Give the full reading.

7 13/16 in

The whole part is 7 in and the counted fraction is 13/16 in.

Textbook Chapter 4 pp.88–96

Learner PDF page 3 • W02-L10-Q11

08 Complete 3/8 = ?/24. Enter the numerator.

9

Multiply both parts by 3: 3/8 = 9/24.

Textbook Chapter 4 pp.88–96

Learner PDF page 4 • W02-L10-Q13

09 Convert 5 7/12 to an improper fraction over 12. Enter its numerator.

67

5 × 12 + 7 = 67, giving 67/12.

Textbook Chapter 4 pp.88–96

Learner PDF page 4 • W02-L10-Q15

10 How many sixteenth-inch intervals represent 5/8 in?

10 intervals

5/8 = 10/16, so there are ten small intervals.

Textbook Chapter 4 pp.88–96

Fraction foundations checkpoint

Day 07 • 2026-11-06 • 15 questions

Learner PDF page 4 • W02-L10-Q17

11 The pictured rectangle’s shorter side is 7¼ in. Rewrite it as an improper fraction over 4: enter the numerator.

29

7 × 4 + 1 = 29, so the width is 29/4 in.

Textbook Chapter 4 pp.88–96

Learner PDF page 5 • W02-L10-Q18

12 A learner simplifies 12/16 to 6/16. Which correction is needed?

D. Divide both numerator and denominator by the same factor

Dividing both parts by 4 gives the correct simplest form, 3/4.

Textbook Chapter 4 pp.88–96

Learner PDF page 5 • W02-L10-Q19

13 Which dimension is shortest?

D. 3 5/8 in

In sixteenths, the fractional parts are 10, 12, 14 and 13; 10/16 is smallest.

Textbook Chapter 4 pp.88–96

Learner PDF page 5 • W02-L10-Q21

14 A tape mark sits 7 sixteenth-inch spaces past the 9 in line. What is the reading?

9 7/16 in

Each space is 1/16 in. 7 spaces = 7/16 in. The reading is 9 7/16 in.

Textbook Chapter 4 pp.88–96

Learner PDF page 6 • W02-L10-Q23

15 A learner records 9 1/2 in and another records 9 4/8 in. What should the check conclude?

A. The two readings are equal

1/2 = 4/8, and both readings have the same whole-inch part.

Textbook Chapter 4 pp.88–96

Adding fractional lengths

Day 08 • 2026-11-09 • 15 questions

Learner PDF page 1 • W03-L01-Q01

01 Before checking a fractional-length sum with the Numeracy Hub calculator, what should you write?

D. A common denominator and the intended sum

Writing equivalent fractions and the intended sum makes a calculator entry error easier to detect.

Textbook Chapter 4 pp.91–99

Learner PDF page 2 • W03-L01-Q02

02 Which fraction is equivalent to 3/4 with denominator 8?

C. 6/8

Multiply numerator and denominator by 2: 3/4 = 6/8.

Textbook Chapter 4 pp.91–99

Learner PDF page 2 • W03-L01-Q03

03 Which statement correctly describes 1/2 + 1/4?

B. 2/4 + 1/4 = 3/4

The parts must have equal size. One half is two quarters.

Textbook Chapter 4 pp.91–99

Learner PDF page 2 • W03-L01-Q04

04 What is the least common denominator of 1/6 and 1/4?

C. 12

12 is the smallest positive number divisible by both 6 and 4.

Textbook Chapter 4 pp.91–99

Learner PDF page 3 • W03-L01-Q05

05 A total of 9/8 in is written as a mixed number. Which is correct?

B. 1 1/8 in

Eight eighths make one whole, leaving one eighth.

Textbook Chapter 4 pp.91–99

Adding fractional lengths

Day 08 • 2026-11-09 • 15 questions

Learner PDF page 3 • W03-L01-Q09

06 The Shape Canvas rectangle shows a long side of 18 1/2 in (18.5 in) and a short side of 7 1/4 in (7.25 in). What is one long side plus one short side?

25 3/4 in

18 1/2 + 7 1/4 = 18 2/4 + 7 1/4 = 25 3/4 in.

Textbook Chapter 4 pp.91–99

Learner PDF page 4 • W03-L01-Q10

07 Using the pictured 7 1/4 in short side, what is the total length of two matching short pieces? Add the two lengths.

14 1/2 in

7 1/4 + 7 1/4 = 14 2/4 = 14 1/2 in.

Textbook Chapter 4 pp.91–99

Learner PDF page 4 • W03-L01-Q11

08 Add 3/8 in and 1/4 in.

5/8 in

Convert 1/4 to 2/8, then add: 3/8 + 2/8 = 5/8 in.

Textbook Chapter 4 pp.91–99

Learner PDF page 4 • W03-L01-Q12

09 Add 4 5/8 in and 2 3/8 in.

7 in

5/8 + 3/8 = 1; 4 + 2 + 1 = 7 in.

Textbook Chapter 4 pp.91–99

Learner PDF page 5 • W03-L01-Q13

10 Add 1/3 in and 1/6 in.

1/2 in

1/3 = 2/6; 2/6 + 1/6 = 3/6 = 1/2 in.

Textbook Chapter 4 pp.91–99

Adding fractional lengths

Day 08 • 2026-11-09 • 15 questions

Learner PDF page 5 • W03-L01-Q17

11 A trim run has pieces 12 3/4 in and 8 5/8 in long. Ignoring joints and kerf, what is their combined length?

21 3/8 in

3/4 = 6/8; 6/8 + 5/8 = 1 3/8. The total is 21 3/8 in.

Textbook Chapter 4 pp.91–99

Learner PDF page 5 • W03-L01-Q18

12 A cut list needs 6 7/8 in, 3 1/4 in and 2 3/8 in. What total length is listed?

12 1/2 in

Convert to eighths: 7/8 + 2/8 + 3/8 = 1 1/2. Add the 11 whole inches to get 12 1/2 in.

Textbook Chapter 4 pp.91–99

Learner PDF page 6 • W03-L01-Q19

13 A learner writes 3/8 + 1/8 = 4/16. What is the correction?

B. Keep the denominator 8; the sum is 4/8 = 1/2

The denominator names the size of each part. Adding eighths does not change them into sixteenths.

Textbook Chapter 4 pp.91–99

Learner PDF page 6 • W03-L01-Q20

14 A measurement is increased from 9 1/4 in by 5/8 in. What is the new measurement?

9 7/8 in

1/4 = 2/8; 2/8 + 5/8 = 7/8. The new measurement is 9 7/8 in.

Textbook Chapter 4 pp.91–99

Learner PDF page 6 • W03-L01-Q21

15 You calculated 4 7/8 + 5 1/8 = 9 in. Which step was missed?

B. Regrouping 8/8 as 1 whole

The fractional parts sum to 1 whole; 4 + 5 + 1 = 10 in.

Textbook Chapter 4 pp.91–99

Subtracting and regrouping fractions

Day 08 • 2026-11-09 • 15 questions

Learner PDF page 1 • W03-L02-Q01

01 After finding a remaining length on paper, how can the Numeracy Hub calculator check it?

B. Add the removed length and remainder to recover the original

Subtraction is checked by reversing it with addition.

Textbook Chapter 4 pp.97–99

Learner PDF page 2 • W03-L02-Q02

02 To subtract 3/4 in from 5/8 in, which equivalent fraction should replace 3/4?

B. 6/8

Multiply numerator and denominator of 3/4 by 2.

Textbook Chapter 4 pp.97–99

Learner PDF page 2 • W03-L02-Q03

03 Which regrouping is correct for 6 1/4 in?

C. 5 5/4 in

One whole is 4/4; moving it to 1/4 produces 5/4 while reducing the whole number to 5.

Textbook Chapter 4 pp.97–99

Learner PDF page 2 • W03-L02-Q04

04 A board is 20 in long and a piece is removed. Which operation finds the remainder?

C. Original length minus removed length

The remaining part is what is left after the cut piece is taken away.

Textbook Chapter 4 pp.97–99

Learner PDF page 3 • W03-L02-Q05

05 When subtracting 1/4 from 3/8, what denominator can both fractions use?

A. 8

1/4 becomes 2/8, allowing subtraction of equal-size parts.

Textbook Chapter 4 pp.97–99

Subtracting and regrouping fractions

Day 08 • 2026-11-09 • 15 questions

Learner PDF page 3 • W03-L02-Q09

06 The pictured long side is 18 1/2 in and the short side is 7 1/4 in. How much longer is the long side?

11 1/4 in

18 2/4 − 7 1/4 = 11 1/4 in.

Textbook Chapter 4 pp.97–99

Learner PDF page 4 • W03-L02-Q10

07 Treat the pictured 18 1/2 in long side as stock. Remove a 9 3/4 in piece, with no kerf for this example. What remains?

8 3/4 in

Regroup 18 1/2 as 17 6/4. Subtract 9 3/4 to get 8 3/4 in.

Textbook Chapter 4 pp.97–99

Learner PDF page 4 • W03-L02-Q11

08 Subtract 1/4 in from 7/8 in.

5/8 in

1/4 = 2/8; 7/8 − 2/8 = 5/8 in.

Textbook Chapter 4 pp.97–99

Learner PDF page 4 • W03-L02-Q12

09 Find 8 1/4 in − 3 5/8 in.

4 5/8 in

8 1/4 = 7 10/8; subtract 3 5/8 to get 4 5/8 in.

Textbook Chapter 4 pp.97–99

Learner PDF page 5 • W03-L02-Q13

10 Find 10 in − 2 3/16 in.

7 13/16 in

Regroup 10 as 9 16/16; the result is 7 13/16 in.

Textbook Chapter 4 pp.97–99

Subtracting and regrouping fractions

Day 08 • 2026-11-09 • 15 questions

Learner PDF page 5 • W03-L02-Q17

11 A 36 1/2 in strip contains a 14 7/8 in part. Ignoring kerf, find the remaining length.

21 5/8 in

36 1/2 = 35 12/8; the remainder is 21 5/8 in.

Textbook Chapter 4 pp.97–99

Learner PDF page 5 • W03-L02-Q18

12 An overall dimension is 24 1/4 in. Two known sections are 8 1/2 in and 7 3/4 in. Find the missing section.

8 in

The known sections total 16 1/4 in; 24 1/4 − 16 1/4 = 8 in.

Textbook Chapter 4 pp.97–99

Learner PDF page 5 • W03-L02-Q19

13 A learner writes 6 1/4 − 2 3/4 = 4 2/4. What is the correct answer?

B. 3 1/2 in

Regroup 6 1/4 as 5 5/4; 5 5/4 − 2 3/4 = 3 2/4 = 3 1/2.

Textbook Chapter 4 pp.97–99

Learner PDF page 6 • W03-L02-Q20

14 A 17 3/8 in piece must be shortened to 15 7/8 in. How much must be removed?

1 1/2 in

17 3/8 = 16 11/8; subtract 15 7/8 to get 1 4/8 = 1 1/2 in.

Textbook Chapter 4 pp.97–99

Learner PDF page 6 • W03-L02-Q21

15 Your remainder is 11 3/8 in after removing 4 5/8 in. Which original length confirms the result?

A. 16 in

11 3/8 + 4 5/8 = 15 + 8/8 = 16 in.

Textbook Chapter 4 pp.97–99

Repeated pieces and equal groups

Day 09 • 2026-11-10 • 15 questions

Learner PDF page 1 • W03-L03-Q01

01 Which entry checks four equal lengths in the Numeracy Hub calculator?

B. 4 × one piece length

Multiplication combines equal groups. Write the intended expression before entering it.

Textbook Chapter 4 pp.100–107

Learner PDF page 1 • W03-L03-Q02

02 Which operation finds the midpoint of a 17 1/2 in board?

C. Divide 17 1/2 by 2

The midpoint splits the total length into two equal lengths.

Textbook Chapter 4 pp.100–107

Learner PDF page 2 • W03-L03-Q03

03 What is the improper-fraction form of 2 3/4?

D. 11/4

2 × 4 + 3 = 11, keeping denominator 4.

Textbook Chapter 4 pp.100–107

Learner PDF page 2 • W03-L03-Q04

04 To divide by 3/4, which equivalent operation is correct?

D. Multiply by 4/3

Dividing by a nonzero fraction is multiplying by its reciprocal.

Textbook Chapter 4 pp.100–107

Learner PDF page 2 • W03-L03-Q05

05 A length of 12 in is divided into three equal sections. What does the answer describe?

A. The length of one section

The total length divided by the number of equal groups gives the length in each group.

Textbook Chapter 4 pp.100–107

Repeated pieces and equal groups

Day 09 • 2026-11-10 • 15 questions

Learner PDF page 3 • W03-L03-Q09

06 The pictured short side is 7 1/4 in (7.25 in). Find the total length of three matching short pieces.

21 3/4 in

3 × 7 1/4 = 3 × 29/4 = 87/4 = 21 3/4 in.

Textbook Chapter 4 pp.100–107

Learner PDF page 4 • W03-L03-Q10

07 The pictured long side is 18 1/2 in (18.5 in). Where is its midpoint measured from one end?

9 1/4 in

18 1/2 ÷ 2 = 37/4 = 9 1/4 in.

Textbook Chapter 4 pp.100–107

Learner PDF page 4 • W03-L03-Q11

08 Find the total length of four pieces each 2 3/8 in long.

9 1/2 in

4 × 19/8 = 76/8 = 9 1/2 in.

Textbook Chapter 4 pp.100–107

Learner PDF page 4 • W03-L03-Q12

09 Split 15 3/4 in into three equal lengths. Find one length.

5 1/4 in

63/4 ÷ 3 = 21/4 = 5 1/4 in.

Textbook Chapter 4 pp.100–107

Learner PDF page 5 • W03-L03-Q13

10 Ignoring kerf, how many 3/4 in spacers fit exactly in 9 in?

12 spacers

9 × 4/3 = 12 spacers.

Textbook Chapter 4 pp.100–107

Repeated pieces and equal groups

Day 09 • 2026-11-10 • 15 questions

Learner PDF page 5 • W03-L03-Q17

11 Eight equal blocking pieces are each 4 1/4 in long. Ignoring kerf, what total length is required?

34 in

8 × 17/4 = 34 in.

Textbook Chapter 4 pp.100–107

Learner PDF page 5 • W03-L03-Q18

12 A 30 in stock strip is divided into pieces 2 1/2 in long, ignoring kerf. How many full pieces fit?

12 pieces

30 ÷ 2.5 = 12 pieces.

Textbook Chapter 4 pp.100–107

Learner PDF page 6 • W03-L03-Q19

13 A learner calculates 3 × 2 1/2 as 6 1/2. What went wrong?

C. The 1/2 in fractional part was not multiplied by 3

Three lots of 2 1/2 are 6 + 3/2 = 7 1/2 in.

Textbook Chapter 4 pp.100–107

Learner PDF page 6 • W03-L03-Q20

14 Four equal gaps fill a total free space of 5 1/2 in. What is one gap?

1 3/8 in

11/2 ÷ 4 = 11/8 = 1 3/8 in.

Textbook Chapter 4 pp.100–107

Learner PDF page 6 • W03-L03-Q21

15 A centre mark is 6 7/8 in from either end of a strip. Find the full strip length.

13 3/4 in

Two equal halves total 13 3/4 in.

Textbook Chapter 4 pp.100–107

Fraction operations: check and retry

Day 09 • 2026-11-10 • 15 questions

Learner PDF page 1 • W03-L04-Q01

01 Add 1/4 in and 3/8 in. Give a fraction in inches.

5/8 in

Rename 1/4 as 2/8. Then 2/8 + 3/8 = 5/8 in.

Textbook Chapter 4 pp.97–107

Learner PDF page 1 • W03-L04-Q02

02 Which operation finds an offcut from a known stock length and used length?

D. Subtraction

The offcut is the original stock minus the length used, accounting for kerf if stated.

Textbook Chapter 4 pp.97–107

Learner PDF page 1 • W03-L04-Q03

03 Which operation finds the total for five identical pieces?

A. Multiplication

Multiply the length of one piece by five.

Textbook Chapter 4 pp.97–107

Learner PDF page 2 • W03-L04-Q04

04 Which operation finds each of four equal gaps in a known total free space?

D. Division by 4

Equal sharing divides the total free space by the number of gaps.

Textbook Chapter 4 pp.97–107

Learner PDF page 2 • W03-L04-Q05

05 Which fraction is the lowest-term form of 6/8?

C. 3/4

Divide numerator and denominator by their common factor 2.

Textbook Chapter 4 pp.97–107

Fraction operations: check and retry

Day 09 • 2026-11-10 • 15 questions

Learner PDF page 3 • W03-L04-Q09

06 Use the pictured 18 1/2 in long side and 7 1/4 in short side. Find their sum in inches.

25 3/4 in

Equivalent quarters give 18 2/4 + 7 1/4 = 25 3/4 in.

Textbook Chapter 4 pp.97–107

Learner PDF page 4 • W03-L04-Q10

07 Use the pictured 18 1/2 in long side and 7 1/4 in short side. Find their difference in inches.

11 1/4 in

18 2/4 − 7 1/4 = 11 1/4 in.

Textbook Chapter 4 pp.97–107

Learner PDF page 4 • W03-L04-Q11

08 Evaluate 2 3/8 + 1 7/8.

4 1/4

10/8 = 1 1/4, so 2 + 1 + 1 1/4 = 4 1/4.

Textbook Chapter 4 pp.97–107

Learner PDF page 4 • W03-L04-Q12

09 Evaluate 7 1/4 − 2 5/8.

4 5/8

Regroup 7 2/8 as 6 10/8; the result is 4 5/8.

Textbook Chapter 4 pp.97–107

Learner PDF page 5 • W03-L04-Q13

10 Evaluate 3 × 1 5/6.

5 1/2

3 × 11/6 = 33/6 = 5 1/2.

Textbook Chapter 4 pp.97–107

Fraction operations: check and retry

Day 09 • 2026-11-10 • 15 questions

Learner PDF page 5 • W03-L04-Q17

11 A learner writes 1/2 + 1/3 = 2/5. Which correction is valid?

B. 3/6 + 2/6 = 5/6

Fractions need common-size parts before addition; the common denominator is 6.

Textbook Chapter 4 pp.97–107

Learner PDF page 5 • W03-L04-Q18

12 A 28 in strip needs three 6 1/2 in pieces. Ignore kerf. Find the offcut.

8 1/2 in

The three pieces use 19 1/2 in; 28 − 19 1/2 = 8 1/2 in.

Textbook Chapter 4 pp.97–107

Learner PDF page 5 • W03-L04-Q19

13 A learner answers 2 for 4 ÷ 1/2. Which explanation corrects the work?

A. Eight halves fit into 4, so the answer is 8

Dividing by 1/2 counts half-size groups; each whole contains two halves.

Textbook Chapter 4 pp.97–107

Learner PDF page 6 • W03-L04-Q20

14 A missing section plus 4 7/8 in must total 13 1/4 in. Find the missing section.

8 3/8 in

13 1/4 − 4 7/8 = 8 3/8 in; adding back 4 7/8 gives 13 1/4 in.

Textbook Chapter 4 pp.97–107

Learner PDF page 6 • W03-L04-Q21

15 A 15 1/4 in board has a 5 7/8 in piece removed. Ignore kerf. What length remains? Check by adding back.

9 3/8 in

Rename 15 1/4 as 15 2/8, then regroup 1 whole: 14 10/8. 14 10/8 − 5 7/8 = 9 3/8 in. Check: 9 3/8 + 5 7/8 = 15 2/8 = 15 1/4 in.

Textbook Chapter 4 pp.97–107

Fraction addition and subtraction: job practice

Day 10 • 2026-11-16 • 15 questions

Learner PDF page 1 • W04-L01-Q01

01 When using the Numeracy Hub calculator to check a mixed-inch sum, which habit matters most?

D. Enter each mixed number with the correct fraction grouping

Incorrect grouping can change a mixed number into a different value.

Textbook sections 4.3–4.6 pp.92–99

Learner PDF page 2 • W04-L01-Q02

02 Which form correctly regroups 9 1/8 for subtracting 5/8?

B. 8 9/8

Move one whole from 9 into the fraction: 8/8 + 1/8 = 9/8.

Textbook sections 4.3–4.6 pp.92–99

Learner PDF page 2 • W04-L01-Q03

03 What is the least common denominator for 3/8 and 5/16?

C. 16

Sixteenths can represent both fractions without changing their values.

Textbook sections 4.3–4.6 pp.92–99

Learner PDF page 2 • W04-L01-Q04

04 A frame part is lengthened by an allowance. Which operation applies the allowance?

B. Addition

An added allowance increases the required length.

Textbook sections 4.3–4.6 pp.92–99

Learner PDF page 3 • W04-L01-Q05

05 A piece is shortened from an existing length to a specified final length. Which calculation gives the amount removed?

C. Existing length minus final length

The removed amount is the difference between the old and new measurements.

Textbook sections 4.3–4.6 pp.92–99

Fraction addition and subtraction: job practice

Day 10 • 2026-11-16 • 15 questions

Learner PDF page 3 • W04-L01-Q09

06 The pictured long side is 18 1/2 in and short side 7 1/4 in. A joint requires both lengths plus 3/8 in allowance. Find the total.

26 1/8 in

18 4/8 + 7 2/8 + 3/8 = 26 1/8 in.

Textbook sections 4.3–4.6 pp.92–99

Learner PDF page 4 • W04-L01-Q10

07 The pictured long side is 18 1/2 in. Subtract the 7 1/4 in short side and a 1/8 in clearance. What length remains?

11 1/8 in

18 1/2 − 7 1/4 = 11 1/4; subtracting 1/8 leaves 11 1/8 in.

Textbook sections 4.3–4.6 pp.92–99

Learner PDF page 4 • W04-L01-Q11

08 Add 14 7/16 in and 8 5/8 in.

23 1/16 in

5/8 = 10/16; the fractional sum is 17/16, so the total is 23 1/16 in.

Textbook sections 4.3–4.6 pp.92–99

Learner PDF page 4 • W04-L01-Q12

09 Subtract 11 13/16 in from 27 1/4 in.

15 7/16 in

27 1/4 = 26 20/16; subtracting gives 15 7/16 in.

Textbook sections 4.3–4.6 pp.92–99

Learner PDF page 5 • W04-L01-Q13

10 Add 6 3/8 in, 4 7/16 in and 2 1/8 in.

12 15/16 in

The fractions are 6/16 + 7/16 + 2/16 = 15/16; the total is 12 15/16 in.

Textbook sections 4.3–4.6 pp.92–99

Fraction addition and subtraction: job practice

Day 10 • 2026-11-16 • 15 questions

Learner PDF page 5 • W04-L01-Q17

11 A jamb assembly uses 28 3/4 in and 19 7/8 in parts plus a 1/2 in allowance. Find the listed total length.

49 1/8 in

Convert to eighths: 6/8 + 7/8 + 4/8 = 2 1/8. Add 47 whole inches to get 49 1/8 in.

Textbook sections 4.3–4.6 pp.92–99

Learner PDF page 5 • W04-L01-Q18

12 A 48 in stock piece provides a 20 5/8 in piece and a 17 7/16 in piece. Ignore kerf. Find the offcut.

9 15/16 in

The used lengths total 38 1/16 in; the offcut is 9 15/16 in.

Textbook sections 4.3–4.6 pp.92–99

Learner PDF page 5 • W04-L01-Q19

13 A learner writes 10 1/4 − 4 3/4 = 6 1/2. Which is the corrected answer?

C. 5 1/2 in

Regroup 10 1/4 as 9 5/4; 9 5/4 − 4 3/4 = 5 2/4.

Textbook sections 4.3–4.6 pp.92–99

Learner PDF page 6 • W04-L01-Q20

14 A 15 3/8 in opening must have 1/16 in clearance at each end of a fitted piece. Find the piece length.

15 1/4 in

Two clearances total 1/8 in; 15 3/8 − 1/8 = 15 1/4 in.

Textbook sections 4.3–4.6 pp.92–99

Learner PDF page 6 • W04-L01-Q21

15 Which equation checks a remaining length of 7 3/8 in after 12 5/8 in is removed?

C. 7 3/8 + 12 5/8 = 20 in

The remainder and removed length must add back to the original stock length.

Textbook sections 4.3–4.6 pp.92–99

Fraction cut lists, centre lines and kerf

Day 10 • 2026-11-16 • 15 questions

Learner PDF page 1 • W04-L02-Q01

01 What should be entered in Numeracy Hub to check total saw loss?

A. Number of actual cuts × kerf per cut

Each actual cut removes one kerf width. The number of cuts must be stated or determined from the cut plan.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Learner PDF page 2 • W04-L02-Q02

02 Four pieces are cut one at a time from a longer board, leaving an offcut. How many cuts are made?

A. 4

Each of the four pieces is separated by one cut; an offcut remains after the fourth cut.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Learner PDF page 2 • W04-L02-Q03

03 A stock length is split into four pieces using only the three internal separating cuts. How many kerfs are counted?

B. 3

The problem explicitly specifies three separating cuts; count actual cuts, not automatically the number of pieces.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Learner PDF page 2 • W04-L02-Q04

04 Which expression finds the centre of a 19 1/4 in length?

C. 19 1/4 ÷ 2

The centre divides the length into equal halves.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Learner PDF page 3 • W04-L02-Q05

05 Which expression gives stock used for three 5 1/2 in pieces and three 1/8 in kerfs?

A. 3 × 5 1/2 + 3 × 1/8

Both the finished pieces and saw loss consume stock length.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Fraction cut lists, centre lines and kerf

Day 10 • 2026-11-16 • 15 questions

Learner PDF page 3 • W04-L02-Q09

06 The pictured short side is 7 1/4 in. Cut three pieces of that length from stock, leaving an offcut. Each of the three cuts removes 1/8 in. Find total stock used.

22 1/8 in

The pieces total 21 3/4 in and kerfs total 3/8 in; stock used is 22 1/8 in.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Learner PDF page 4 • W04-L02-Q10

07 Find the centre of the pictured 18 1/2 in long side, measured from one end.

9 1/4 in

18 1/2 ÷ 2 = 9 1/4 in.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Learner PDF page 4 • W04-L02-Q11

08 Five cuts each remove 3/32 in. Find the total saw loss.

15/32 in

5 × 3/32 = 15/32 in.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Learner PDF page 4 • W04-L02-Q12

09 Four pieces are each 8 3/8 in. Find their total finished length before kerf.

33 1/2 in

4 × 67/8 = 268/8 = 33 1/2 in.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Learner PDF page 5 • W04-L02-Q13

10 A 36 in stock piece provides four 8 3/8 in pieces and four 1/8 in kerfs. Find the offcut.

2 in

Finished lengths total 33 1/2 in; kerfs total 1/2 in. The offcut is 2 in.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Fraction cut lists, centre lines and kerf

Day 10 • 2026-11-16 • 15 questions

Learner PDF page 5 • W04-L02-Q17

11 A 48 in strip supplies five 9 1/4 in pieces with five 1/8 in cuts, leaving an offcut. Find the offcut.

1 1/8 in

Pieces use 46 1/4 in; kerfs use 5/8 in; 48 − 46 7/8 = 1 1/8 in.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Learner PDF page 5 • W04-L02-Q18

12 A 60 in strip supplies six 9 3/4 in pieces with six 3/32 in cuts. Find the length left over.

15/16 in

Pieces use 58 1/2 in; kerfs use 9/16 in; the offcut is 15/16 in.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Learner PDF page 6 • W04-L02-Q19

13 A learner subtracts the kerf from the finished-piece total to find required stock. What is the error?

B. Kerf consumes stock and must be added to the finished total

Stock must provide both the finished parts and the material removed by the saw.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Learner PDF page 6 • W04-L02-Q20

14 A centre line lies 11 5/16 in from each end. Find the total uncut length.

22 5/8 in

Twice 11 5/16 is 22 10/16 = 22 5/8 in.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Learner PDF page 6 • W04-L02-Q21

15 Which result proves that 24 in stock cannot supply three 8 in pieces with three 1/8 in cuts?

C. The requirement is 24 3/8 in

The pieces alone use all 24 in; adding 3/8 in saw loss exceeds the stock.

Textbook sections 4.7 pp.100–107; fraction worksheets pp.108–110

Decimals: place value and operations

Day 11 • 2026-11-17 • 15 questions

Learner PDF page 1 • W04-L03-Q01

01 When reading a metric rule like the Numeracy Hub example, what should a recorded measurement include?

B. A numerical value and its unit

A reading such as 37.5 mm identifies both magnitude and measurement unit.

Textbook sections 5.1–5.5 pp.114–119

Learner PDF page 2 • W04-L03-Q02

02 In 46.372, which digit is in the hundredths place?

C. 7

The first digit after the decimal is tenths, the second hundredths, and the third thousandths.

Textbook sections 5.1–5.5 pp.114–119

Learner PDF page 2 • W04-L03-Q03

03 Which two values are equal?

C. 5.20 and 5.2

A trailing zero after the final decimal digit does not change the value.

Textbook sections 5.1–5.5 pp.114–119

Learner PDF page 2 • W04-L03-Q04

04 How should numbers be lined up for decimal addition?

D. Align the decimal points

Aligning decimal points places ones, tenths and hundredths in the correct columns.

Textbook sections 5.1–5.5 pp.114–119

Learner PDF page 3 • W04-L03-Q05

05 Which comparison is correct?

C. 0.8 > 0.75

Writing 0.8 as 0.80 makes the comparison clear: 80 hundredths exceeds 75 hundredths.

Textbook sections 5.1–5.5 pp.114–119

Decimals: place value and operations

Day 11 • 2026-11-17 • 15 questions

Learner PDF page 3 • W04-L03-Q09

06 Read the pictured 1200 mm and 800 mm sides. Two offcuts of 12.5 mm each are added to the 1200 mm length for a quantity total. Find that total.

1225 mm

1200 + 25 = 1225 mm.

Textbook sections 5.1–5.5 pp.114–119

Learner PDF page 4 • W04-L03-Q10

07 From the pictured 800 mm dimension, subtract a 25.75 mm allowance. What dimension remains?

774.25 mm

800.00 − 25.75 = 774.25 mm.

Textbook sections 5.1–5.5 pp.114–119

Learner PDF page 4 • W04-L03-Q11

08 Add 12.45 mm and 7.8 mm.

20.25 mm

Align decimal places: 12.45 + 7.80 = 20.25 mm.

Textbook sections 5.1–5.5 pp.114–119

Learner PDF page 4 • W04-L03-Q12

09 Find 50 mm − 18.65 mm.

31.35 mm

Write 50.00 − 18.65 = 31.35 mm.

Textbook sections 5.1–5.5 pp.114–119

Learner PDF page 5 • W04-L03-Q13

10 Four identical spacers are 2.35 mm thick. Find their combined thickness.

9.4 mm

4 × 2.35 = 9.40 mm.

Textbook sections 5.1–5.5 pp.114–119

Decimals: place value and operations

Day 11 • 2026-11-17 • 15 questions

Learner PDF page 5 • W04-L03-Q17

11 A measured length is 125.75 mm and the target is 124.5 mm. How much material must be removed?

1.25 mm

125.75 − 124.50 = 1.25 mm.

Textbook sections 5.1–5.5 pp.114–119

Learner PDF page 5 • W04-L03-Q18

12 A cut list contains 16.8 mm, 22.35 mm and 9.75 mm. Find the total length.

48.9 mm

16.80 + 22.35 + 9.75 = 48.90 mm.

Textbook sections 5.1–5.5 pp.114–119

Learner PDF page 5 • W04-L03-Q19

13 A calculator displays 346 for 3.46 × 10. Which correction is needed?

C. The product should be 34.6

Multiplying by 10 changes the value by one place, not two.

Textbook sections 5.1–5.5 pp.114–119

Learner PDF page 6 • W04-L03-Q20

14 A 45.6 mm gap is shared equally between four spaces. Find one space.

11.4 mm

45.6 ÷ 4 = 11.4 mm. Check: 4 × 11.4 = 45.6.

Textbook sections 5.1–5.5 pp.114–119

Learner PDF page 6 • W04-L03-Q21

15 Which written check supports 7.25 + 2.6 = 9.85?

A. 9.85 − 2.6 = 7.25

Subtraction reverses addition and confirms the total.

Textbook sections 5.1–5.5 pp.114–119

Decimals, tape rounding and costs

Day 11 • 2026-11-17 • 15 questions

Learner PDF page 1 • W04-L04-Q01

01 Which operation converts 3/8 to a decimal using the Numeracy Hub calculator?

D. 3 ÷ 8

A fraction represents its numerator divided by its denominator.

Textbook sections 5.6–5.7 pp.120–132

Learner PDF page 1 • W04-L04-Q02

02 What is 0.25 as a fraction in lowest terms?

B. 1/4

0.25 = 25/100 = 1/4.

Textbook sections 5.6–5.7 pp.120–132

Learner PDF page 2 • W04-L04-Q03

03 To round a decimal inch measurement to the nearest 1/16 in, which method works?

C. Multiply by 16, round to a whole number, then divide by 16

This measures the value in sixteenth-inch steps, then selects the nearest step.

Textbook sections 5.6–5.7 pp.120–132

Learner PDF page 2 • W04-L04-Q04

04 Which amount rounds to $12.35 to the nearest cent?

B. $12.346

The third decimal digit is 6, so the hundredths digit rounds up.

Textbook sections 5.6–5.7 pp.120–132

Learner PDF page 2 • W04-L04-Q05

05 What does $4.50 per metre mean?

A. Each metre costs $4.50

A unit price is multiplied by the required number of units.

Textbook sections 5.6–5.7 pp.120–132

Decimals, tape rounding and costs

Day 11 • 2026-11-17 • 15 questions

Learner PDF page 3 • W04-L04-Q09

06 The pictured long side is 18.5 in. What is its fractional part as a decimal number of inches?

0.5 in

18.5 − 18 = 0.5 in, equal to 1/2 in.

Textbook sections 5.6–5.7 pp.120–132

Learner PDF page 4 • W04-L04-Q10

07 The pictured 18.5 in long side is extended by 0.375 in. Which mixed-number measurement gives the new length?

D. 18 7/8 in

18.5 + 0.375 = 18.875 in. Since 0.875 = 7/8, the new length is 18 7/8 in.

Textbook sections 5.6–5.7 pp.120–132

Learner PDF page 4 • W04-L04-Q11

08 Which decimal inch measurement is equal to 7/16 in?

C. 0.4375 in

Divide the numerator by the denominator: 7 ÷ 16 = 0.4375 in.

Textbook sections 5.6–5.7 pp.120–132

Learner PDF page 5 • W04-L04-Q12

09 Round 6.43 in to the nearest 1/16 in. Enter the resulting length as a fraction or decimal.

6 7/16 in

6.43 × 16 = 102.88; round to 103 sixteenths. 103/16 = 6 7/16 in.

Textbook sections 5.6–5.7 pp.120–132

Learner PDF page 5 • W04-L04-Q13

10 Calculate the cost of 4.5 m of trim at $3.20 per metre.

14.40 $

4.5 × $3.20 = $14.40.

Textbook sections 5.6–5.7 pp.120–132

Decimals, tape rounding and costs

Day 11 • 2026-11-17 • 15 questions

Learner PDF page 5 • W04-L04-Q17

11 A piece must be at least 5.14 in long. Round upward to the next 1/8 in increment.

5 1/4 in

5.14 × 8 = 41.12 eighths. Round upward to 42 eighths = 5 1/4 in.

Textbook sections 5.6–5.7 pp.120–132

Learner PDF page 5 • W04-L04-Q18

12 A purchase includes 3.75 m at $4.80/m and two brackets at $2.35 each. Find the total before tax.

22.70 $

The trim costs $18.00 and brackets cost $4.70; total $22.70.

Textbook sections 5.6–5.7 pp.120–132

Learner PDF page 6 • W04-L04-Q19

13 A learner changes 3/4 in to 0.34 in. What should the calculator check show?

B. 3 ÷ 4 = 0.75 in

The numerator and denominator are not decimal digits to place side by side.

Textbook sections 5.6–5.7 pp.120–132

Learner PDF page 6 • W04-L04-Q20

14 A $40 budget is used to buy 6 m of material at $5.25/m. How much remains?

8.50 $

The material costs $31.50; $40.00 − $31.50 = $8.50.

Textbook sections 5.6–5.7 pp.120–132

Learner PDF page 6 • W04-L04-Q21

15 Which reason supports keeping full calculator precision until a cost total is complete?

D. Rounding every intermediate value can accumulate error

Perform the specified calculations first, then round the final monetary amount to cents.

Textbook sections 5.6–5.7 pp.120–132

Percent: part of a whole

Day 12 • 2026-11-18 • 15 questions

Learner PDF page 1 • W04-L05-Q01

01 Which expression checks 15% of 240 using the Numeracy Hub calculator?

D. 0.15 × 240

Percent means per hundred, so 15% becomes 0.15 before multiplication.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Learner PDF page 1 • W04-L05-Q02

02 Which decimal equals 8%?

A. 0.08

8% = 8/100 = 0.08.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Learner PDF page 2 • W04-L05-Q03

03 Which percentage equals 0.35?

A. 35%

Multiply the decimal by 100 to express it as a percentage.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Learner PDF page 2 • W04-L05-Q04

04 Which fraction equals 25%?

B. 1/4

25/100 reduces to 1/4.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Learner PDF page 2 • W04-L05-Q05

05 What is the whole when finding what percentage 18 accepted pieces represent out of 24 inspected pieces?

C. 24 pieces

The full inspected group is the denominator of the part-to-whole comparison.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Percent: part of a whole

Day 12 • 2026-11-18 • 15 questions

Learner PDF page 3 • W04-L05-Q09

06 The pictured long side is 1200 mm. Find 25% of that length.

300 mm

25% is one quarter; 1200 ÷ 4 = 300 mm.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Learner PDF page 4 • W04-L05-Q10

07 The pictured short side is 800 mm. Find 15% of that length.

120 mm

15% = 0.15; 0.15 × 800 = 120 mm.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Learner PDF page 4 • W04-L05-Q11

08 Find 12% of 250 pieces.

30 pieces

0.12 × 250 = 30 pieces.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Learner PDF page 4 • W04-L05-Q12

09 Express 3/5 as a percentage.

60 %

3 ÷ 5 = 0.6; 0.6 × 100 = 60%.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Learner PDF page 5 • W04-L05-Q13

10 What percentage of 80 boards is 20 boards?

25 %

20/80 = 0.25; 0.25 × 100 = 25%.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Percent: part of a whole

Day 12 • 2026-11-18 • 15 questions

Learner PDF page 5 • W04-L05-Q17

11 A batch contains 120 pieces, and 18 need repair. What percentage need repair?

15 %

18 ÷ 120 = 0.15; 15% of the batch needs repair.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Learner PDF page 5 • W04-L05-Q18

12 A material total is 360 kg. One component makes up 35% of the total. Find its mass.

126 kg

0.35 × 360 = 126 kg.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Learner PDF page 5 • W04-L05-Q19

13 A learner finds 6% of 200 as 1200. What correction is needed?

D. Use 0.06 × 200 = 12

The percentage must be divided by 100 before it acts as a multiplier.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Learner PDF page 6 • W04-L05-Q20

14 A crew completes 42 of 60 planned tasks. What percentage is complete?

70 %

42 ÷ 60 = 0.7; 70% is complete.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Learner PDF page 6 • W04-L05-Q21

15 Which answer is reasonable for 10% of 95 m?

A. 9.5 m

10% is one tenth; divide 95 by 10.

Textbook sections 6.1–6.3 pp.138–142; calculator percent p.55

Waste allowance and purchase quantities

Day 12 • 2026-11-18 • 15 questions

Learner PDF page 1 • W04-L06-Q01

01 Which Numeracy Hub calculator expression applies a 10% allowance once to a net quantity of 80?

A. 80 × 1.10

1.10 includes 100% of the net quantity plus 10% extra.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Learner PDF page 2 • W04-L06-Q02

02 Which multiplier applies a 15% discount?

A. 0.85

After a 15% discount, 85% of the original price remains.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Learner PDF page 2 • W04-L06-Q03

03 An order needs 7.2 packs, and packs cannot be split. How many must be purchased?

A. 8 packs

A purchase must cover the requirement, so round upward to the next whole pack.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Learner PDF page 2 • W04-L06-Q04

04 An allowance is stated as 8% of the net quantity. What is the percentage base?

B. The net quantity before allowance

Apply the stated allowance to the specified net requirement, once.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Learner PDF page 3 • W04-L06-Q05

05 Which expression gives the extra material only for 12% allowance on 50 m?

A. 50 × 0.12

Multiplying by 0.12 gives the extra 12%, not the complete order.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Waste allowance and purchase quantities

Day 12 • 2026-11-18 • 15 questions

Learner PDF page 3 • W04-L06-Q09

06 Treat the pictured 1200 mm long side as the net material length. Apply a 10% allowance once. Find the order length.

1320 mm

10% of 1200 is 120 mm; 1200 + 120 = 1320 mm.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Learner PDF page 4 • W04-L06-Q10

07 Treat the pictured 800 mm short side as the net length. Find the extra length only for a 15% allowance.

120 mm

0.15 × 800 = 120 mm extra.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Learner PDF page 4 • W04-L06-Q11

08 A net quantity is 45 m. Add 8% allowance. Find the adjusted quantity.

48.6 m

The extra is 3.6 m; the adjusted amount is 48.6 m.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Learner PDF page 4 • W04-L06-Q12

09 A $160 tool is discounted by 15%. Find the price before tax.

136.00 $

The discount is $24; $160 − $24 = $136.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Learner PDF page 5 • W04-L06-Q13

10 A requirement is 34.5 m. Each roll supplies 8 m. How many whole rolls must be bought?

5 rolls

34.5 ÷ 8 = 4.3125; buy five whole rolls.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Waste allowance and purchase quantities

Day 12 • 2026-11-18 • 15 questions

Learner PDF page 5 • W04-L06-Q17

11 A job needs 92 m net plus 10% allowance. Stock is sold in 12 m bundles. How many whole bundles are required?

9 bundles

Adjusted need is 101.2 m. 101.2 ÷ 12 = 8.433...; buy 9 bundles.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Learner PDF page 5 • W04-L06-Q18

12 A job needs 38 kg net plus 15% allowance. Bags contain 20 kg each. How many whole bags are needed?

3 bags

38 × 1.15 = 43.7 kg; 43.7 ÷ 20 = 2.185, so buy 3 bags.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Learner PDF page 6 • W04-L06-Q19

13 A requirement including allowance is 10.01 boxes. Which purchase covers it?

D. 11 boxes

Even a small amount above ten boxes requires an eleventh whole box.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Learner PDF page 6 • W04-L06-Q20

14 The net cost of material is $240. A 10% discount is applied once. Find the amount payable before tax.

216.00 $

The discount is $24; the payable amount is $216.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Learner PDF page 6 • W04-L06-Q21

15 Which check helps detect an incorrect allowance calculation?

A. The adjusted quantity should exceed the positive net quantity

Adding a positive allowance increases the material requirement while keeping its unit.

Textbook sections 6.4 pp.142–143; review sections 5.7 pp.124–132

Ratios, rates and proportions

Day 13 • 2026-11-19 • 15 questions

Learner PDF page 1 • W04-L07-Q01

01 Before checking a ratio in the Numeracy Hub calculator, what must be made consistent?

D. The units of both quantities

A ratio of like quantities should use common units before simplification.

Textbook sections 6.5–6.7 pp.144–148

Learner PDF page 1 • W04-L07-Q02

02 Which ratio is 12:18 in lowest whole-number terms?

A. 2:3

Divide both terms by their greatest common factor, 6.

Textbook sections 6.5–6.7 pp.144–148

Learner PDF page 2 • W04-L07-Q03

03 Which ratio is equivalent to 3:5?

C. 6:10

Multiplying both terms by 2 preserves the ratio.

Textbook sections 6.5–6.7 pp.144–148

Learner PDF page 2 • W04-L07-Q04

04 What does a rate of 4 boards per hour describe?

A. Four boards completed in each hour at that rate

A rate compares different units: boards and hours.

Textbook sections 6.5–6.7 pp.144–148

Learner PDF page 2 • W04-L07-Q05

05 If the number of identical frames doubles while the material per frame stays fixed, what happens to total material?

D. It doubles

A constant quantity per frame creates a proportional relationship.

Textbook sections 6.5–6.7 pp.144–148

Ratios, rates and proportions

Day 13 • 2026-11-19 • 15 questions

Learner PDF page 3 • W04-L07-Q09

06 The pictured rectangle is 1200 mm by 800 mm. Its long-to-short ratio is 3:2. If the short side becomes 1000 mm at the same ratio, find the long side.

1500 mm

The long side is 3/2 of the short side: 1000 × 3/2 = 1500 mm.

Textbook sections 6.5–6.7 pp.144–148

Learner PDF page 4 • W04-L07-Q10

07 The pictured rectangle is 1200 mm by 800 mm. Scale both sides by 1.5. Find the new short side.

1200 mm

800 × 1.5 = 1200 mm. Both dimensions must use the same scale factor.

Textbook sections 6.5–6.7 pp.144–148

Learner PDF page 4 • W04-L07-Q11

08 Three identical packs contain 72 screws altogether. How many screws are in one pack?

24 screws

72 ÷ 3 = 24 screws per pack.

Textbook sections 6.5–6.7 pp.144–148

Learner PDF page 4 • W04-L07-Q12

09 A worker completes 18 equal pieces in 3 hours. At the same rate, how many pieces are completed in 5 hours?

30 pieces

The rate is 6 pieces per hour; 6 × 5 = 30 pieces.

Textbook sections 6.5–6.7 pp.144–148

Learner PDF page 5 • W04-L07-Q13

10 A mix uses A:B = 2:3. If 8 kg of A is used, how much B is required?

12 kg

The A quantity is four lots of 2 kg; B is four lots of 3 kg = 12 kg.

Textbook sections 6.5–6.7 pp.144–148

Ratios, rates and proportions

Day 13 • 2026-11-19 • 15 questions

Learner PDF page 5 • W04-L07-Q17

11 Four identical frames require 18 m of material. At the same rate, how many metres are needed for ten frames?

45 m

Each frame requires 4.5 m; ten frames require 45 m.

Textbook sections 6.5–6.7 pp.144–148

Learner PDF page 5 • W04-L07-Q18

12 A mix is A:B = 3:2. The total mix is 35 kg. Find the mass of B.

14 kg

Five parts total 35 kg, so each part is 7 kg. B uses two parts: 14 kg.

Textbook sections 6.5–6.7 pp.144–148

Learner PDF page 5 • W04-L07-Q19

13 A learner simplifies 1200 mm:800 mm to 2:3. What is the correction?

C. The long-to-short ratio is 3:2

1200:800 divided by 400 is 3:2. Reversing the terms changes the comparison.

Textbook sections 6.5–6.7 pp.144–148

Learner PDF page 6 • W04-L07-Q20

14 At a constant rate, 7 m of trim costs $29.40. What do 12 m cost?

50.40 $

The unit price is $4.20/m; 12 m cost $50.40.

Textbook sections 6.5–6.7 pp.144–148

Learner PDF page 6 • W04-L07-Q21

15 Which check supports a proportional material estimate?

C. The quantity per item remains the same

Direct proportion preserves the unit rate when the job conditions are unchanged.

Textbook sections 6.5–6.7 pp.144–148

Scale drawings and weekly check

Day 13 • 2026-11-19 • 15 questions

Learner PDF page 1 • W04-L08-Q01

01 Before using a Numeracy Hub calculator result for a scale drawing, what must be checked?

C. The scale direction and consistent length units

Multiplying or dividing in the wrong direction can create a plausible but incorrect length.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Learner PDF page 2 • W04-L08-Q02

02 At a scale of 1:50, what does 1 mm on the drawing represent?

B. 50 mm in reality

The ratio compares drawing length to real length using the same unit.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Learner PDF page 2 • W04-L08-Q03

03 Which operation changes a drawing length into a real length at 1:20 scale?

D. Multiply by 20

Real length is 20 times drawing length at this reduction scale.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Learner PDF page 2 • W04-L08-Q04

04 Which operation finds drawing length from a real length at 1:100 scale?

C. Divide by 100

Drawing length is one hundredth of real length when units match.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Learner PDF page 3 • W04-L08-Q05

05 Which information should control a cut made from a drawing?

B. The stated dimension and units

Screen size and printing can change; use stated dimensions unless a verified scale measurement is specifically required.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Scale drawings and weekly check

Day 13 • 2026-11-19 • 15 questions

Learner PDF page 3 • W04-L08-Q09

06 The pictured real floor is 7.2 m long. At 1:50 scale, what is its drawing length in millimetres? Use 1 m = 1000 mm.

144 mm

7.2 m = 7200 mm; 7200 ÷ 50 = 144 mm.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Learner PDF page 4 • W04-L08-Q10

07 The pictured real floor is 3.6 m wide. At 1:100 scale, what is its drawing width in millimetres? Use 1 m = 1000 mm.

36 mm

3.6 m = 3600 mm; 3600 ÷ 100 = 36 mm.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Learner PDF page 4 • W04-L08-Q11

08 A drawing length is 48 mm at 1:25 scale. Find the real length in millimetres.

1200 mm

48 × 25 = 1200 mm.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Learner PDF page 4 • W04-L08-Q12

09 A real length is 1800 mm. Find its drawing length at 1:20 scale.

90 mm

1800 ÷ 20 = 90 mm.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Learner PDF page 5 • W04-L08-Q13

10 A 1:50 drawing shows two adjoining lengths of 24 mm and 36 mm. Find their combined real length in metres. Use 1000 mm = 1 m.

3 m

Drawing total is 60 mm; real total is 3000 mm = 3 m.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Scale drawings and weekly check

Day 13 • 2026-11-19 • 15 questions

Learner PDF page 5 • W04-L08-Q17

11 A 1:50 drawing has an overall line of 120 mm and a known section of 45 mm. Find the remaining real length in millimetres.

3750 mm

The drawing remainder is 75 mm; 75 × 50 = 3750 mm.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Learner PDF page 5 • W04-L08-Q18

12 A cut list requires four real lengths represented by 32 mm each on a 1:25 drawing. Ignore kerf. Find the total real length in metres. Use 1000 mm = 1 m.

3.2 m

Each part is 800 mm; four total 3200 mm = 3.2 m.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Learner PDF page 6 • W04-L08-Q19

13 A learner multiplies a 2000 mm real length by 50 to create its 1:50 drawing length. What correction is needed?

C. Divide 2000 by 50 to get 40 mm

A reduced drawing is smaller than the real object, so divide by the scale denominator.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Learner PDF page 6 • W04-L08-Q20

14 A 1:20 drawing shows a net material length of 85 mm. The real order needs 10% allowance. Find the order length in millimetres.

1870 mm

Real net length is 1700 mm; multiplying once by 1.10 gives 1870 mm.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Scale drawings and weekly check

Day 13 • 2026-11-19 • 15 questions

Learner PDF page 6 • W04-L08-Q21

15 A 1:50 drawing is displayed at an unknown screen zoom. Which method gives a reliable real dimension?

B. Use its printed dimension label and the stated unit

An unknown zoom breaks the physical scale; stated dimensions retain their meaning.

Textbook sections 6.8 pp.148–150; review Chapter 4–6

Measurement systems and readable dimensions

Day 14 • 2026-11-24 • 15 questions

Learner PDF page 1 • W05-L01-Q01

01 Which unit is most useful for recording the thickness of a plywood sheet?

D. Millimetres

Thickness is a length; millimetres give a practical scale for a sheet a few millimetres thick.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Learner PDF page 2 • W05-L01-Q02

02 Read the Shape Canvas rectangle. Express its 1200 mm length in centimetres.

120 cm

There are 10 mm in 1 cm, so 1200 ÷ 10 = 120 cm.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Learner PDF page 3 • W05-L01-Q03

03 The Numeracy Hub measurement illustration supports a ruler reading. Where should a measured length start?

B. At the zero graduation

Align the object with the zero graduation, or subtract the starting reading from the ending reading.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Learner PDF page 4 • W05-L01-Q04

04 Read the Shape Canvas rectangle. Express its 800 mm width in metres.

0.8 m

There are 1000 mm in 1 m, so 800 mm = 0.8 m.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Learner PDF page 4 • W05-L01-Q05

05 Which statement describes a dimension marked 6 ft 4 in?

D. Six feet plus four inches

The feet and inches are separate quantities to add after converting to a common unit.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Measurement systems and readable dimensions

Day 14 • 2026-11-24 • 15 questions

Learner PDF page 5 • W05-L01-Q09

06 Convert 2.75 m to millimetres.

2,750 mm

Multiply metres by 1000: 2.75 × 1000 = 2750 mm.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Learner PDF page 5 • W05-L01-Q10

07 Convert 845 mm to centimetres.

84.5 cm

Divide millimetres by 10: 845 ÷ 10 = 84.5 cm.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Learner PDF page 5 • W05-L01-Q11

08 Convert 4 ft 7 in to inches.

55 in

Four feet is 48 in; 48 + 7 = 55 in.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Learner PDF page 5 • W05-L01-Q12

09 Convert 96 in to feet.

8 ft

There are 12 in per foot: 96 ÷ 12 = 8 ft.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Learner PDF page 6 • W05-L01-Q13

10 A tape shows 2 m and an additional 35 cm. What is the total in centimetres?

235 cm

Convert 2 m to 200 cm, then add 35 cm to obtain 235 cm.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Measurement systems and readable dimensions

Day 14 • 2026-11-24 • 15 questions

Learner PDF page 6 • W05-L01-Q17

11 A learner writes 3 ft 6 in = 3.6 ft. What is the correct decimal-foot value?

D. 3.5 ft

Six inches is 6 ÷ 12 = 0.5 ft, so the total is 3.5 ft.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Learner PDF page 6 • W05-L01-Q18

12 Two pieces measure 1.2 m and 650 mm. What is their combined length in millimetres?

1,850 mm

Convert 1.2 m to 1200 mm; 1200 + 650 = 1850 mm.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Learner PDF page 7 • W05-L01-Q19

13 A 2.4 m stock length is cut by 85 cm. Ignore saw kerf. How many millimetres remain?

1,550 mm

2400 mm − 850 mm = 1550 mm.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Learner PDF page 7 • W05-L01-Q20

14 Which conversion has the correct size and direction?

A. 650 mm = 0.65 m

A metre is larger than a millimetre, so the numerical value becomes smaller: 650 ÷ 1000 = 0.65.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Learner PDF page 8 • W05-L01-Q21

15 A rectangle is 18.5 in long as shown in Shape Canvas. How much longer is it than a 12 in piece?

6.5 in

Read the dimension label, then subtract: 18.5 − 12 = 6.5 in.

Textbook sections 7.1–7.2 pp.164–166; sections 4.2 pp.89–90

Exact conversion factors and mixed-unit practice

Day 14 • 2026-11-24 • 15 questions

Learner PDF page 1 • W05-L02-Q01

01 Which exact relationship should you use to convert inches to millimetres?

D. 1 in = 25.4 mm

The exact conversion factor is 25.4 millimetres per inch.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Learner PDF page 2 • W05-L02-Q02

02 The Shape Canvas rectangle is 18.5 in long. Convert the length to millimetres using 25.4 mm per inch.

469.9 mm

18.5 × 25.4 = 469.9 mm.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Learner PDF page 3 • W05-L02-Q03

03 Using Numeracy Hub to check a conversion, which operation converts millimetres to inches?

B. Divide by 25.4

Dividing mm by mm per inch leaves inches.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Learner PDF page 4 • W05-L02-Q04

04 The Shape Canvas rectangle is 7.25 in wide. Convert its width to millimetres.

184.15 mm

7.25 × 25.4 = 184.15 mm.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Learner PDF page 4 • W05-L02-Q05

05 When a mixed-unit calculation asks for a final answer to the nearest millimetre, when should you round?

D. After the calculation is complete

Keep conversion precision through the working and round only the final requested quantity.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Exact conversion factors and mixed-unit practice

Day 14 • 2026-11-24 • 15 questions

Learner PDF page 5 • W05-L02-Q09

06 Convert 3.5 in to millimetres.

88.9 mm

3.5 × 25.4 = 88.9 mm.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Learner PDF page 5 • W05-L02-Q10

07 Convert 508 mm to inches.

20 in

508 ÷ 25.4 = 20 in.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Learner PDF page 5 • W05-L02-Q11

08 Convert 6 ft to metres.

1.8288 m

6 ft = 72 in; 72 × 25.4 = 1828.8 mm = 1.8288 m.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Learner PDF page 5 • W05-L02-Q12

09 Convert 2 ft 6 in to millimetres.

762 mm

2 ft 6 in = 30 in, and 30 × 25.4 = 762 mm.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Learner PDF page 6 • W05-L02-Q13

10 Convert 1200 mm to inches. Round to 2 decimal places.

47.24 in

1200 ÷ 25.4 = 47.244094… in, which rounds to 47.24 in.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Exact conversion factors and mixed-unit practice

Day 14 • 2026-11-24 • 15 questions

Learner PDF page 6 • W05-L02-Q17

11 A rail is 900 mm long and another is 12 in long. What is their combined length in millimetres?

1,204.8 mm

12 in = 304.8 mm; 900 + 304.8 = 1204.8 mm.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Learner PDF page 7 • W05-L02-Q18

12 A student uses 25 mm per inch for an exact conversion. What should change?

C. Use 25.4 mm per inch

25 mm per inch is a rough estimate; exact work uses 25.4 mm per inch.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Learner PDF page 7 • W05-L02-Q19

13 A 1200 mm board needs a 20 in piece cut from it. Ignore kerf. What length remains in millimetres?

692 mm

20 in = 508 mm; 1200 − 508 = 692 mm.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Learner PDF page 7 • W05-L02-Q20

14 A drawing gives 3 ft 4 in. Record the length to the nearest millimetre.

1,016 mm

3 ft 4 in = 40 in, and 40 × 25.4 = 1016 mm.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Learner PDF page 8 • W05-L02-Q21

15 Which reverse check confirms that 762 mm equals 30 in?

D. 30 × 25.4 = 762

Multiplying the inch value by the conversion factor returns the original millimetre value.

Textbook sections 7.3 pp.167–169; conversion factors pp.362–363

Reading and substituting into formulas

Day 15 • 2026-11-25 • 15 questions

Learner PDF page 1 • W05-L03-Q01

01 In A = L × W for a rectangle, what does W represent?

D. Width

The letters name quantities: A is area, L is length, and W is width.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Learner PDF page 2 • W05-L03-Q02

02 Use A = L × W for the Shape Canvas rectangle measuring 1200 mm by 800 mm. Find its area.

960,000 mm²

A = 1200 × 800 = 960000 mm². The product of two lengths has square units.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Learner PDF page 3 • W05-L03-Q03

03 The Numeracy Hub calculator checks P = 2(L + W). For L = 8 and W = 3, which entry keeps the formula’s grouping?

B. 2 × (8 + 3)

The brackets combine length and width before doubling the sum.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Learner PDF page 4 • W05-L03-Q04

04 Use P = 2(L + W) for the 7.2 m by 3.6 m Shape Canvas floor. Find P.

21.6 m

P = 2(7.2 + 3.6) = 2 × 10.8 = 21.6 m.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Learner PDF page 4 • W05-L03-Q05

05 What unit belongs to area when L and W are both in metres?

D. m²

Metres multiplied by metres gives square metres.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Reading and substituting into formulas

Day 15 • 2026-11-25 • 15 questions

Learner PDF page 4 • W05-L03-Q09

06 Use A = L × W when L = 4.5 m and W = 2.2 m.

9.9 m²

A = 4.5 × 2.2 = 9.9 m².

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Learner PDF page 5 • W05-L03-Q10

07 Use P = 2(L + W) when L = 650 mm and W = 280 mm.

1,860 mm

P = 2(650 + 280) = 2 × 930 = 1860 mm.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Learner PDF page 5 • W05-L03-Q11

08 Use V = L × W × H when L = 2 m, W = 1.5 m and H = 0.4 m.

1.2 m³

V = 2 × 1.5 × 0.4 = 1.2 m³.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Learner PDF page 5 • W05-L03-Q12

09 Use A = b × h / 2 with b = 8 m and perpendicular h = 3 m.

12 m²

A = 8 × 3 ÷ 2 = 12 m².

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Learner PDF page 5 • W05-L03-Q13

10 Use T = n × l + w with n = 6, l = 350 mm and w = 20 mm. Find T.

2,120 mm

T = 6 × 350 + 20 = 2100 + 20 = 2120 mm.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Reading and substituting into formulas

Day 15 • 2026-11-25 • 15 questions

Learner PDF page 6 • W05-L03-Q17

11 A student substitutes into 2(L + W) as 2L + W. What has been missed?

D. The width must also be doubled

The distributive form is 2L + 2W, because 2 multiplies both terms in the brackets.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Learner PDF page 6 • W05-L03-Q18

12 A floor is 3.5 m by 2800 mm. Use A = L × W and give the area in m².

9.8 m²

Convert 2800 mm to 2.8 m; A = 3.5 × 2.8 = 9.8 m².

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Learner PDF page 6 • W05-L03-Q19

13 Which result can be a perimeter?

B. 18.4 m

Perimeter is a distance around a boundary, so it uses linear units.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Learner PDF page 6 • W05-L03-Q20

14 Material cost is C = A × p. If A = 12.5 m² and p = $8 per m², find C.

$100.00

C = 12.5 × 8 = $100. The square-metre units cancel with price per square metre.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Learner PDF page 7 • W05-L03-Q21

15 After calculating a formula, which check is strongest?

D. Check the substituted values, units and approximate size

Correct input, consistent units and a reasonable size provide separate checks on the result.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Rearranging formulas and checking backwards

Day 15 • 2026-11-25 • 15 questions

Learner PDF page 1 • W05-L04-Q01

01 Rearrange A = L × W to find W.

D. W = A ÷ L

Divide both sides by L to leave W on its own.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Learner PDF page 2 • W05-L04-Q02

02 A new floor keeps the 3.6 m width shown in Shape Canvas but has area 43.2 m². Find its new length.

12 m

L = A ÷ W = 43.2 ÷ 3.6 = 12 m. Use the shown width, not the old drawing’s length.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Learner PDF page 3 • W05-L04-Q03

03 Which Numeracy Hub calculator entry correctly finds W from P = 2(L + W), P = 30 and L = 9?

B. 30 ÷ 2 − 9

First undo the multiplication by 2, then subtract L: W = P/2 − L.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Learner PDF page 4 • W05-L04-Q04

04 A rectangle’s area is 960000 mm². Its width is 800 mm. Which rearrangement gives the length shown in Shape Canvas?

A. L = 960000 ÷ 800

Length equals area divided by width. This gives 1200 mm.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Learner PDF page 4 • W05-L04-Q05

05 How should a calculated missing dimension be checked?

D. Substitute it into the original formula

Returning to the original equation checks that the missing value reproduces the given result.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Rearranging formulas and checking backwards

Day 15 • 2026-11-25 • 15 questions

Learner PDF page 5 • W05-L04-Q09

06 A rectangle has area 18 m² and width 2.4 m. Find its length.

7.5 m

L = A/W = 18 ÷ 2.4 = 7.5 m.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Learner PDF page 5 • W05-L04-Q10

07 P = 2(L + W), P = 38 m and L = 12 m. Find W.

7 m

W = P/2 − L = 19 − 12 = 7 m.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Learner PDF page 5 • W05-L04-Q11

08 A = b × h / 2, A = 24 m² and b = 8 m. Find h.

6 m

Multiply by 2, then divide by b: h = 2A/b = 48 ÷ 8 = 6 m.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Learner PDF page 5 • W05-L04-Q12

09 V = L × W × H, V = 4.8 m³, L = 4 m and W = 2 m. Find H.

0.6 m

H = V/(LW) = 4.8 ÷ 8 = 0.6 m.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Learner PDF page 6 • W05-L04-Q13

10 T = 5l + 25, with T = 1825 mm. Find l in millimetres.

360 mm

Subtract 25 before dividing by 5: l = 1800 ÷ 5 = 360 mm.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Rearranging formulas and checking backwards

Day 15 • 2026-11-25 • 15 questions

Learner PDF page 6 • W05-L04-Q17

11 A student finds W = P − 2L from P = 2(L + W). Why is that incomplete?

D. P − 2L equals 2W, so divide by 2

Expanding gives P = 2L + 2W. Subtracting 2L leaves twice the width.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Learner PDF page 6 • W05-L04-Q18

12 A 6 m by W rectangle has area 33 m². Find W, then use multiplication to check.

5.5 m

W = 5.5 m. Substitution gives 6 × 5.5 = 33 m².

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Learner PDF page 7 • W05-L04-Q19

13 A missing width calculated from positive area and positive length is negative. What should happen next?

B. Recheck the rearrangement and arithmetic

A physical width cannot be negative here, so the result signals an error to investigate.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Learner PDF page 7 • W05-L04-Q20

14 A rectangular frame has P = 17.6 m and W = 3.4 m. Find L.

5.4 m

L = 8.8 − 3.4 = 5.4 m; 2(5.4 + 3.4) = 17.6 m.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Learner PDF page 7 • W05-L04-Q21

15 For V = LWH, which expression isolates H correctly?

D. H = V ÷ (L × W)

Divide by the entire product LW. The brackets preserve the intended divisor.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Hidden dimensions: square roots and formula checks

Day 16 • 2026-11-26 • 15 questions

Learner PDF page 1 • W05-L05-Q01

01 For a square with area A, which operation finds the positive side length?

D. Take the square root of A

The formula $A = s^2$ is undone by taking the positive square root.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Learner PDF page 2 • W05-L05-Q02

02 Use the 900 mm side shown by Shape Canvas to calculate the square’s area.

810,000 mm²

$A = 900^2 = 810000$ mm². A later square-root check should recover 900 mm.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Learner PDF page 3 • W05-L05-Q03

03 When Numeracy Hub shows a square-root key, which check confirms a computed positive root?

B. Square the result

If $s = \sqrt{A}$, then s^2 must reproduce A.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Learner PDF page 4 • W05-L05-Q04

04 A new rectangle keeps the 3.6 m width shown in Shape Canvas but has area 32.4 m². Find its new length.

9 m

A rectangle’s missing length is $A/W = 32.4 \div 3.6 = 9$ m. The old 7.2 m length belongs to the pictured floor, not this new rectangle.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Learner PDF page 4 • W05-L05-Q05

05 A rectangle has known area and one known side. Which method finds the other side?

D. Divide area by the known side

Only a square has equal side lengths. A general rectangle uses $A = LW$ and division.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Hidden dimensions: square roots and formula checks

Day 16 • 2026-11-26 • 15 questions

Learner PDF page 5 • W05-L05-Q09

06 A square panel has area 14400 mm². Find its side length.

120 mm

s = √14400 = 120 mm, and 120² = 14400.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Learner PDF page 5 • W05-L05-Q10

07 A square floor has area 30.25 m². Find its side length.

5.5 m

s = √30.25 = 5.5 m.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Learner PDF page 5 • W05-L05-Q11

08 A square has area 2 m². Find its side to 3 decimal places.

1.414 m

√2 = 1.414213... m, rounded to 1.414 m.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Learner PDF page 5 • W05-L05-Q12

09 A rectangular panel has area 4.32 m² and length 2.4 m. Find its width.

1.8 m

W = A/L = 4.32 ÷ 2.4 = 1.8 m.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Learner PDF page 6 • W05-L05-Q13

10 A circular area formula is A = pi × r². Use pi = 3.14 and A = 31400 mm² to find r.

100 mm

First isolate r² = A/pi = 10000, then r = √10000 = 100 mm.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Hidden dimensions: square roots and formula checks

Day 16 • 2026-11-26 • 15 questions

Learner PDF page 6 • W05-L05-Q17

11 A student calculates √64 as 32. Which check exposes the error?

D. $32^2 = 1024$, not 64

A square root must square back to the original quantity. The correct positive root is 8.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Learner PDF page 6 • W05-L05-Q18

12 A square's proposed side is 1.8 m. Calculate its area as a substitution check.

3.24 m²

$1.8 \times 1.8 = 3.24$ m².

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Learner PDF page 7 • W05-L05-Q19

13 For r = √(A/pi), which calculator grouping is correct?

B. Take the square root of the whole quotient A/pi

The division must be inside the square-root operation.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Learner PDF page 7 • W05-L05-Q20

14 A square area is 9 m². Its side will be recorded in millimetres. Find the side in mm.

3,000 mm

√9 = 3 m. Convert 3 m to 3000 mm.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Learner PDF page 7 • W05-L05-Q21

15 A square-root calculation gives a side of about 4.47 m for area 20 m². Which estimate supports it?

D. The side is between 4 m and 5 m

$4^2 = 16$ and $5^2 = 25$; since 20 is between those areas, its root is between 4 and 5.

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Perimeter, frame lengths and material totals

Day 17 • 2026-11-27 • 15 questions

Learner PDF page 1 • W05-L06-Q01

01 What does perimeter measure?

D. The length around the outside boundary

Perimeter follows the entire outside edge once.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Learner PDF page 2 • W05-L06-Q02

02 Find the perimeter of the 1200 mm by 800 mm Shape Canvas rectangle.

4,000 mm

P = 2(1200 + 800) = 4000 mm.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Learner PDF page 3 • W05-L06-Q03

03 In the Numeracy Hub area/perimeter illustration, should an internal dividing line be counted in the outside perimeter?

B. No; only outside boundary segments count

Internal partitions do not belong to the outside boundary.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Learner PDF page 4 • W05-L06-Q04

04 Find the perimeter of the 900 mm side-length Shape Canvas square.

3,600 mm

A square has four equal sides, so P = 4 × 900 = 3600 mm.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Learner PDF page 4 • W05-L06-Q05

05 Which unit is appropriate for a frame’s perimeter?

D. mm

A frame length is a linear measurement.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Perimeter, frame lengths and material totals

Day 17 • 2026-11-27 • 15 questions

Learner PDF page 4 • W05-L06-Q09

06 Find the perimeter of a rectangle 4.8 m long and 2.5 m wide.

14.6 m

P = 2 × 7.3 = 14.6 m.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Learner PDF page 5 • W05-L06-Q10

07 A rectangle has perimeter 34 m and length 11 m. Find its width.

6 m

W = P/2 − L = 17 − 11 = 6 m.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Learner PDF page 5 • W05-L06-Q11

08 A square has perimeter 5200 mm. Find one side.

1,300 mm

Four equal sides total 5200 mm, so each side is 1300 mm.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Learner PDF page 5 • W05-L06-Q12

09 A five-sided outline has sides 900, 600, 450, 700 and 550 mm. Find its perimeter.

3,200 mm

Sum every outside segment once: 3200 mm.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Learner PDF page 5 • W05-L06-Q13

10 A rectangular room is 6 m by 4 m. Find the skirting length after excluding a 0.9 m doorway.

19.1 m

The perimeter is 20 m. Subtract the doorway: 20 − 0.9 = 19.1 m.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Perimeter, frame lengths and material totals

Day 17 • 2026-11-27 • 15 questions

Learner PDF page 6 • W05-L06-Q17

11 A learner finds 5 × 3 = 15 m for the perimeter of a 5 m by 3 m rectangle. What went wrong?

D. They used the area operation instead of adding boundary lengths

Multiplication gives area. Perimeter is 2(5 + 3) = 16 m.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Learner PDF page 6 • W05-L06-Q18

12 A 7.2 m by 3.6 m floor needs trim around all four sides. Stock comes in 3 m lengths. Ignoring joins and cuts, what is the minimum whole number of lengths by total length?

8 lengths

Perimeter is 21.6 m. 21.6 ÷ 3 = 7.2, so round up to 8 stock lengths. A cut plan may need more.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Learner PDF page 7 • W05-L06-Q19

13 Two rectangles have the same area. Must their perimeters be the same?

B. No; different proportions can give different perimeters

For example, 2 × 8 and 4 × 4 both have area 16, but perimeters 20 and 16.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Learner PDF page 7 • W05-L06-Q20

14 An L-shaped outline has six boundary lengths: 6, 2, 2, 2, 4 and 4 m. Find its perimeter.

20 m

Walk the outside and add each length once: 20 m.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Perimeter, frame lengths and material totals

Day 17 • 2026-11-27 • 15 questions

Learner PDF page 7 • W05-L06-Q21

15 Which check helps prevent counting a boundary segment twice?

D. Trace the outside from one corner back to the same corner

One continuous trip around the boundary accounts for each segment exactly once.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Circumference, arcs and perimeter consolidation

Day 17 • 2026-11-27 • 15 questions

Learner PDF page 1 • W05-L07-Q01

01 Which formula finds circumference when diameter d is known?

D. C = pi × d

Circumference is pi times the diameter.

Textbook sections 9.2–9.6 pp.222–229

Learner PDF page 2 • W05-L07-Q02

02 The Shape Canvas circle has radius 350 mm. Find its diameter.

700 mm

Diameter is twice the radius: d = 700 mm.

Textbook sections 9.2–9.6 pp.222–229

Learner PDF page 3 • W05-L07-Q03

03 Consider a semicircular panel, separate from the Numeracy Hub rectangle shown. Which pieces make the panel’s full perimeter?

B. Half the circumference plus the diameter

The boundary includes the curved arc and the straight cut edge.

Textbook sections 9.2–9.6 pp.222–229

Learner PDF page 4 • W05-L07-Q04

04 Find the circumference of the Shape Canvas circle with radius 350 mm. Use pi = 3.14.

2,198 mm

C = 2 × 3.14 × 350 = 2198 mm.

Textbook sections 9.2–9.6 pp.222–229

Learner PDF page 4 • W05-L07-Q05

05 What fraction of a circle’s circumference is a 90° arc?

D. One quarter

90 ÷ 360 = 1/4.

Textbook sections 9.2–9.6 pp.222–229

Circumference, arcs and perimeter consolidation

Day 17 • 2026-11-27 • 15 questions

Learner PDF page 5 • W05-L07-Q09

06 A circle has diameter 0.8 m. Find its circumference using pi = 3.14.

2.512 m

C = 3.14 × 0.8 = 2.512 m.

Textbook sections 9.2–9.6 pp.222–229

Learner PDF page 5 • W05-L07-Q10

07 A circular opening has radius 125 mm. Find its circumference using pi = 3.14.

785 mm

C = 2 × 3.14 × 125 = 785 mm.

Textbook sections 9.2–9.6 pp.222–229

Learner PDF page 5 • W05-L07-Q11

08 Find the length of a 90° arc with radius 2 m. Use pi = 3.14.

3.14 m

A quarter of the 12.56 m circumference is 3.14 m.

Textbook sections 9.2–9.6 pp.222–229

Learner PDF page 5 • W05-L07-Q12

09 Find the full perimeter of a semicircular panel with radius 1.5 m. Use pi = 3.14.

7.71 m

Half the circumference is pi r = 4.71 m. Add the 3 m diameter: 7.71 m.

Textbook sections 9.2–9.6 pp.222–229

Learner PDF page 6 • W05-L07-Q13

10 A 120° arc has radius 450 mm. Find its length using pi = 3.14.

942 mm

The arc is one third of 2826 mm, so it is 942 mm.

Textbook sections 9.2–9.6 pp.222–229

Circumference, arcs and perimeter consolidation

Day 17 • 2026-11-27 • 15 questions

Learner PDF page 6 • W05-L07-Q17

11 A student uses C = pi × r when r is the radius. What is missing?

D. A factor of 2

The full circumference is 2pi r, or pi d when diameter is used.

Textbook sections 9.2–9.6 pp.222–229

Learner PDF page 6 • W05-L07-Q18

12 A curved boundary is a semicircle of radius 2 m joined to two straight sides of 3 m each. Find the total of those three boundary pieces. Use pi = 3.14.

12.28 m

The semicircular arc is 6.28 m; add the two stated straight sides to get 12.28 m.

Textbook sections 9.2–9.6 pp.222–229

Learner PDF page 7 • W05-L07-Q19

13 A calculation gives circumference 50 mm for a circle 200 mm in diameter. Which estimate reveals the problem?

B. Circumference should be a little over 600 mm

Circumference is about three times diameter: about 3 × 200 = 600 mm.

Textbook sections 9.2–9.6 pp.222–229

Learner PDF page 7 • W05-L07-Q20

14 A semicircular trim has radius 350 mm as in Shape Canvas. Find only its curved length using pi = 3.14.

1,099 mm

A semicircular arc is half the full circumference: pi r = 1099 mm.

Textbook sections 9.2–9.6 pp.222–229

Learner PDF page 8 • W05-L07-Q21

15 When calculating an arc length, why is angle/360 included?

D. It gives the fraction of the full circle

The central angle determines what fraction of a full 360° turn the arc occupies.

Textbook sections 9.2–9.6 pp.222–229

Area and material coverage

Day 18 • 2026-11-30 • 15 questions

Learner PDF page 1 • W06-L01-Q01

01 What does area measure?

D. The surface covered inside a boundary

Area counts coverage in square units.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Learner PDF page 1 • W06-L01-Q02

02 Find the area of the 7.2 m by 3.6 m Shape Canvas floor.

25.92 m²

A = L × W = 7.2 × 3.6 = 25.92 m².

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Learner PDF page 2 • W06-L01-Q03

03 The Numeracy Hub area/perimeter visual shows a rectangle. Which operation finds area?

B. Multiply length by width

A rectangular area is rows times columns of square units: A = LW.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Learner PDF page 3 • W06-L01-Q04

04 Use the Shape Canvas right triangle with base 4 m and perpendicular height 3 m. Find its area.

6 m²

A triangle is half a rectangle with the same base and height: A = 4 × 3 ÷ 2 = 6 m².

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Learner PDF page 3 • W06-L01-Q05

05 Which height belongs in A = b × h / 2?

D. The perpendicular distance from base to opposite vertex

The triangle formula requires the height perpendicular to the selected base.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Area and material coverage

Day 18 • 2026-11-30 • 15 questions

Learner PDF page 4 • W06-L01-Q09

06 A rectangular panel is 2.4 m by 1.2 m. Find its area.

2.88 m²

A = 2.4 × 1.2 = 2.88 m².

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Learner PDF page 4 • W06-L01-Q10

07 A triangular panel has base 1500 mm and perpendicular height 800 mm. Find its area in mm².

600,000 mm²

A = 1500 × 800 ÷ 2 = 600000 mm².

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Learner PDF page 4 • W06-L01-Q11

08 Convert 750000 mm² to m².

0.75 m²

1 m² = 1000000 mm², so divide by one million to get 0.75 m².

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Learner PDF page 4 • W06-L01-Q12

09 A floor is 4.5 m by 3.2 m. One tile covers 0.36 m². How many tiles cover the area exactly, ignoring cuts?

40 tiles

Floor area is 14.4 m². 14.4 ÷ 0.36 = 40 tiles.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Learner PDF page 5 • W06-L01-Q13

10 A wall is 3200 mm long and 2.4 m high. Find its area in m².

7.68 m²

Convert 3200 mm to 3.2 m; 3.2 × 2.4 = 7.68 m².

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Area and material coverage

Day 18 • 2026-11-30 • 15 questions

Learner PDF page 5 • W06-L01-Q17

11 A student records a 3 m by 4 m floor as 12 m. What is wrong?

D. The area unit must be m²

The numerical area is 12, but multiplying metres by metres gives m².

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Learner PDF page 6 • W06-L01-Q18

12 A floor needs 25.92 m² of coverage. Each sheet covers 2.88 m². How many sheets are required by area, ignoring cuts?

9 sheets

25.92 ÷ 2.88 = 9 sheets by area. Layout and waste are separate checks.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Learner PDF page 6 • W06-L01-Q19

13 A triangle’s base is horizontal. Which labelled measure should be used as its height?

B. A line perpendicular to that base

The altitude, not an arbitrary side, determines the base-times-height area.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Learner PDF page 7 • W06-L01-Q20

14 A room has area 16 m². Add 5% allowance for coverage. Find the total area to plan for.

16.8 m²

Allowance is 0.8 m². Total = 16.8 m².

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Learner PDF page 7 • W06-L01-Q21

15 A calculation divides mm² by 1000 to convert to m². What factor is needed instead?

D. Divide by 1000000

The linear factor 1000 must be squared: 1000² = 1000000.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Composite areas, openings and surface coverage

Day 18 • 2026-11-30 • 15 questions

Learner PDF page 1 • W06-L02-Q01

01 How is the net paintable area of a wall with a window found?

D. Subtract the window area from the whole wall area

An opening removes surface coverage, so subtract its area.

Textbook sections 10.4–10.7 pp.246–254

Learner PDF page 2 • W06-L02-Q02

02 The Shape Canvas floor is 7.2 m by 3.6 m. A 1.2 m by 0.9 m opening is excluded. Find the remaining area.

24.84 m²

Gross area is 25.92 m²; opening area is 1.08 m². Net area is 24.84 m².

Textbook sections 10.4–10.7 pp.246–254

Learner PDF page 3 • W06-L02-Q03

03 The Numeracy Hub area visual supports splitting a composite surface. What must be avoided when adding the parts?

B. Counting overlapping area twice

Parts must be non-overlapping, or any overlap must be subtracted once.

Textbook sections 10.4–10.7 pp.246–254

Learner PDF page 4 • W06-L02-Q04

04 A panel has the Shape Canvas rectangle’s 1200 mm by 800 mm outside dimensions. A 200 mm by 300 mm opening is removed. Find net area in mm².

900,000 mm²

Gross area is 960000 mm²; the opening is 60000 mm². Net area is 900000 mm².

Textbook sections 10.4–10.7 pp.246–254

Learner PDF page 4 • W06-L02-Q05

05 Which units can be subtracted directly from 12 m²?

D. 3 m²

Only like quantities in matching units can be subtracted directly.

Textbook sections 10.4–10.7 pp.246–254

Composite areas, openings and surface coverage

Day 18 • 2026-11-30 • 15 questions

Learner PDF page 5 • W06-L02-Q09

06 A wall is 5 m by 2.4 m and has a 0.9 m by 2.1 m doorway. Find net wall area.

10.11 m²

Gross area is 12 m²; doorway is 1.89 m²; net is 10.11 m².

Textbook sections 10.4–10.7 pp.246–254

Learner PDF page 5 • W06-L02-Q10

07 An L-shaped floor is a 6 m by 5 m rectangle with a 2 m by 3 m corner removed. Find its area.

24 m²

30 − 6 = 24 m².

Textbook sections 10.4–10.7 pp.246–254

Learner PDF page 5 • W06-L02-Q11

08 A gable consists of a 4 m by 2.4 m rectangle topped by a triangle of base 4 m and height 1.2 m. Find total area.

12 m²

Rectangle area is 9.6 m²; triangle area is 2.4 m²; total is 12 m².

Textbook sections 10.4–10.7 pp.246–254

Learner PDF page 5 • W06-L02-Q12

09 A 7 m by 2.7 m wall contains two 1.2 m by 1.5 m windows. Find net area.

15.3 m²

Gross area is 18.9 m². Two windows total 3.6 m²; net is 15.3 m².

Textbook sections 10.4–10.7 pp.246–254

Learner PDF page 6 • W06-L02-Q13

10 A closed rectangular box is 2 m by 1 m by 0.5 m. Use SA = 2(LW + LH + WH). Find surface area.

7 m²

SA = 2(2 + 1 + 0.5) = 7 m².

Textbook sections 10.4–10.7 pp.246–254

Composite areas, openings and surface coverage

Day 18 • 2026-11-30 • 15 questions

Learner PDF page 6 • W06-L02-Q17

11 A student subtracts a window’s 4 m perimeter from 20 m² of wall area. What is wrong?

D. They mixed length with area

Calculate the window’s area in m² before subtracting it from wall area.

Textbook sections 10.4–10.7 pp.246–254

Learner PDF page 6 • W06-L02-Q18

12 A net surface is 18 m². Paint covers 9 m² per litre per coat. Find the theoretical litres for two coats.

4 L

Two coats cover 36 m²; 36 ÷ 9 = 4 L before waste or purchase rounding.

Textbook sections 10.4–10.7 pp.246–254

Learner PDF page 7 • W06-L02-Q19

13 Two rectangles overlap by 2 m². Their separate areas are 12 m² and 9 m². Which expression gives their union area?

B. 12 + 9 − 2

Adding both rectangles counts the overlap twice, so subtract it once.

Textbook sections 10.4–10.7 pp.246–254

Learner PDF page 7 • W06-L02-Q20

14 A net cladding area is 22.4 m². Add 10% waste allowance. Find the planned coverage area.

24.64 m²

22.4 + 2.24 = 24.64 m².

Textbook sections 10.4–10.7 pp.246–254

Composite areas, openings and surface coverage

Day 18 • 2026-11-30 • 15 questions

Learner PDF page 7 • W06-L02-Q21

15 A large rectangle minus a cutout and a sum of non-overlapping parts give different areas. What is the best next step?

D. Check dimensions and whether the parts cover exactly the same region

Equivalent decompositions must agree; a mismatch points to omitted, repeated or incorrect dimensions.

Textbook sections 10.4–10.7 pp.246–254

Volume of boxes and prisms

Day 19 • 2026-12-01 • 15 questions

Learner PDF page 1 • W06-L03-Q01

01 Which formula finds a rectangular prism’s volume?

D. V = L × W × H

Three perpendicular lengths multiply to give cubic units.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Learner PDF page 1 • W06-L03-Q02

02 Find the volume of the Shape Canvas prism measuring 2.4 m by 1.2 m by 0.6 m.

1.728 m³

V = 2.4 × 1.2 × 0.6 = 1.728 m³.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Learner PDF page 2 • W06-L03-Q03

03 Use Numeracy Hub to check a box-volume calculation. Which unit belongs to a volume calculated from dimensions in metres?

B. m³

Metres multiplied three times gives cubic metres.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Learner PDF page 3 • W06-L03-Q04

04 The Shape Canvas floor is 7.2 m by 3.6 m. A slab over it is 0.1 m thick. Find its volume.

2.592 m³

Base area is 25.92 m². Multiply by thickness: 25.92 × 0.1 = 2.592 m³.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Learner PDF page 3 • W06-L03-Q05

05 How many litres are in 1 m³?

D. 1000 L

A litre is 0.001 m³, so 1 m³ contains 1000 L.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Volume of boxes and prisms

Day 19 • 2026-12-01 • 15 questions

Learner PDF page 4 • W06-L03-Q09

06 A box is 3 m long, 2 m wide and 0.75 m high. Find its volume.

4.5 m³

V = 3 × 2 × 0.75 = 4.5 m³.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Learner PDF page 4 • W06-L03-Q10

07 A container’s internal dimensions are 600 mm by 400 mm by 300 mm. Find its capacity in litres.

72 L

Volume is 72000000 mm³. One litre is 1000000 mm³, so capacity is 72 L.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Learner PDF page 4 • W06-L03-Q11

08 A triangular prism has cross-sectional base 1.2 m, perpendicular height 0.8 m, and length 3 m. Find its volume.

1.44 m³

Triangle area = 1.2 × 0.8 ÷ 2 = 0.48 m². Volume = 0.48 × 3 = 1.44 m³.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Learner PDF page 4 • W06-L03-Q12

09 A slab is 5 m by 3 m and 120 mm thick. Find its volume in m³.

1.8 m³

120 mm = 0.12 m; volume = 5 × 3 × 0.12 = 1.8 m³.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Learner PDF page 5 • W06-L03-Q13

10 Convert 2.75 m³ to litres.

2,750 L

2.75 × 1000 = 2750 L.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Volume of boxes and prisms

Day 19 • 2026-12-01 • 15 questions

Learner PDF page 5 • W06-L03-Q17

11 A learner calculates the capacity of a thick-walled tank using outside dimensions. What should be used instead?

D. Internal dimensions

Capacity measures the space available inside; wall material occupies some external volume.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Learner PDF page 6 • W06-L03-Q18

12 Use the Shape Canvas prism’s 1.728 m³ volume. How many litres is that?

1,728 L

1.728 × 1000 = 1728 L.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Learner PDF page 6 • W06-L03-Q19

13 A slab calculation gives 3.6 m³. Add 5% order allowance before delivery-size rounding.

3.78 m³

Allowance = 0.18 m³; total = 3.78 m³.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Learner PDF page 7 • W06-L03-Q20

14 A prism’s length doubles while its cross-section stays unchanged. What happens to volume?

A. It doubles

V = A × length, so volume is directly proportional to length when A is fixed.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Learner PDF page 7 • W06-L03-Q21

15 Which check detects an answer accidentally labelled m² for a box volume?

D. Count the three multiplied length units

Three length dimensions give m × m × m = m³.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Cylinders, capacity and missing heights

Day 19 • 2026-12-01 • 15 questions

Learner PDF page 1 • W06-L04-Q01

01 Which formula finds a cylinder’s volume when radius r and height h are known?

D. V = pi × r² × h

The circular base area is pi r²; multiply by height to obtain volume.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Learner PDF page 2 • W06-L04-Q02

02 The Shape Canvas cylinder has radius 0.5 m and height 1.2 m. Find volume using pi = 3.14.

0.942 m³

V = 3.14 × 0.25 × 1.2 = 0.942 m³.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Learner PDF page 3 • W06-L04-Q03

03 When using Numeracy Hub to calculate a cylinder from diameter, what must be done before squaring?

B. Divide diameter by 2 to get radius

The r in pi r² is radius, which is half the diameter.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Learner PDF page 4 • W06-L04-Q04

04 Use the Shape Canvas cylinder’s volume of 0.942 m³. Express it in litres.

942 L

Multiply cubic metres by 1000: 0.942 m³ = 942 L.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Learner PDF page 4 • W06-L04-Q05

05 Which rearrangement finds cylinder height from volume V and radius r?

D. h = V ÷ (pi × r²)

Divide volume by circular base area to isolate height.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Cylinders, capacity and missing heights

Day 19 • 2026-12-01 • 15 questions

Learner PDF page 5 • W06-L04-Q09

06 A cylinder has radius 0.3 m and height 2 m. Find its volume using pi = 3.14.

0.5652 m³

V = 3.14 × 0.09 × 2 = 0.5652 m³.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Learner PDF page 5 • W06-L04-Q10

07 A tank has internal diameter 1 m and height 1.5 m. Find its capacity in litres using pi = 3.14.

1,177.5 L

Radius is 0.5 m. Volume is 1.1775 m³, or 1177.5 L.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Learner PDF page 5 • W06-L04-Q11

08 A cylinder has volume 3.14 m³ and radius 0.5 m. Find its height using pi = 3.14.

4 m

Base area = 0.785 m². Height = 3.14 ÷ 0.785 = 4 m.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Learner PDF page 5 • W06-L04-Q12

09 A cylindrical tank contains 628 L and has radius 0.4 m. Find the liquid depth using pi = 3.14.

1.25 m

628 L = 0.628 m³. Depth = 0.628 ÷ 0.5024 = 1.25 m.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Learner PDF page 6 • W06-L04-Q13

10 Use 1 US gal = 3.78541 L. Convert 75 L to US gallons, rounding to 2 decimal places.

19.81 US gal

75 ÷ 3.78541 = 19.8129… US gal, rounded to 19.81 US gal.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Cylinders, capacity and missing heights

Day 19 • 2026-12-01 • 15 questions

Learner PDF page 6 • W06-L04-Q17

11 A student enters V/pi × r² to find h. Why should brackets be used?

D. The entire product pi × r² must divide V

The correct expression is V/(pi r²). Without grouping, multiplying by r² after division gives the wrong height.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Learner PDF page 7 • W06-L04-Q18

12 The Shape Canvas cylinder holds 942 L using pi = 3.14. It is half full. How many litres does it contain?

471 L

Half the full capacity is 942 ÷ 2 = 471 L.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Learner PDF page 7 • W06-L04-Q19

13 A height from V/(pi r²) is to be in metres. Which input units are consistent?

B. V in m³ and r in m

m³ divided by m² leaves metres. Other units require explicit conversion first.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Learner PDF page 8 • W06-L04-Q20

14 A rectangular tank with internal dimensions 1.2 m by 0.8 m needs to hold 480 L. Find required liquid depth.

0.5 m

480 L = 0.48 m³. Base area = 0.96 m²; depth = 0.48 ÷ 0.96 = 0.5 m.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Learner PDF page 8 • W06-L04-Q21

15 Why must a gallon conversion name US or Imperial gallons?

D. The two gallon units have different volumes

US and Imperial gallons use different litre factors, so the label matters.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Angles, triangles and drawing interpretation

Day 20 • 2026-12-02 • 15 questions

Learner PDF page 1 • W06-L05-Q01

01 What is the angle marked by the square corner in this Shape Canvas triangle?

90 degrees

The square corner symbol denotes a right angle, which is 90 degrees.

Textbook sections 12.1–12.3 pp.298–300

Learner PDF page 2 • W06-L05-Q02

02 A dimensioned drawing shows 125 degrees. Which angle type is this?

D. Obtuse

An obtuse angle is greater than 90 degrees and less than 180 degrees.

Textbook sections 12.1–12.3 pp.298–300

Learner PDF page 2 • W06-L05-Q03

03 When using the Numeracy Hub measurement diagram, which evidence gives a reliable angle?

C. Its labelled value or angle marker

Screen size and perspective can change. Use stated values and the relevant angle markers.

Textbook sections 12.1–12.3 pp.298–300

Learner PDF page 3 • W06-L05-Q04

04 A full turn is divided into four equal turns. How many degrees is each turn?

90 degrees

Divide 360 degrees by 4 to obtain 90 degrees.

Textbook sections 12.1–12.3 pp.298–300

Learner PDF page 3 • W06-L05-Q05

05 In this right triangle, what is the sum of the two acute angles?

90 degrees

The three interior angles total 180 degrees. Subtract the known right angle: 180 − 90 = 90 degrees.

Textbook sections 12.1–12.3 pp.298–300

Angles, triangles and drawing interpretation

Day 20 • 2026-12-02 • 15 questions

Learner PDF page 3 • W06-L05-Q09

06 Two angles are complementary. One is 34 degrees. Find the other.

56 degrees

Complementary angles total 90 degrees: 90 − 34 = 56 degrees.

Textbook sections 12.1–12.3 pp.298–300

Learner PDF page 4 • W06-L05-Q10

07 A straight line is split into adjacent angles of 118 degrees and x. Find x.

62 degrees

Adjacent angles on a straight line total 180 degrees: 180 − 118 = 62 degrees.

Textbook sections 12.1–12.3 pp.298–300

Learner PDF page 4 • W06-L05-Q11

08 A triangle has angles of 47 degrees and 68 degrees. Find its third angle.

65 degrees

Subtract both known angles from 180: 180 − 47 − 68 = 65 degrees.

Textbook sections 12.1–12.3 pp.298–300

Learner PDF page 4 • W06-L05-Q12

09 One acute angle in a right triangle is 29 degrees. Find the other acute angle.

61 degrees

The two acute angles in a right triangle total 90 degrees: 90 − 29 = 61 degrees.

Textbook sections 12.1–12.3 pp.298–300

Learner PDF page 4 • W06-L05-Q13

10 A triangle has two equal angles of 52 degrees. Find the remaining angle.

76 degrees

The equal angles use 104 degrees. The remaining angle is 180 − 104 = 76 degrees.

Textbook sections 12.1–12.3 pp.298–300

Angles, triangles and drawing interpretation

Day 20 • 2026-12-02 • 15 questions

Learner PDF page 5 • W06-L05-Q17

11 A learner adds 60 and 70 and reports a triangle’s missing angle as 130 degrees. What correction is needed?

A. Subtract 130 from 180

The total of known angles is not the missing angle. 180 − (60 + 70) = 50 degrees.

Textbook sections 12.1–12.3 pp.298–300

Learner PDF page 5 • W06-L05-Q18

12 Which set can be the interior angles of a triangle?

D. 40, 60 and 80 degrees

40 + 60 + 80 = 180 degrees; the other sets have different totals.

Textbook sections 12.1–12.3 pp.298–300

Learner PDF page 5 • W06-L05-Q19

13 Check a proposed right triangle with an acute angle of 36 degrees. What second acute angle makes the total correct?

54 degrees

A right triangle needs 90 degrees in its two acute angles: 90 − 36 = 54 degrees.

Textbook sections 12.1–12.3 pp.298–300

Learner PDF page 6 • W06-L05-Q20

14 You are asked for the angle inside a triangle, but the drawing labels its outside neighbour as 115 degrees. What should you use?

B. 180 − 115

The interior and adjacent exterior angles make a straight angle, so the interior is 65 degrees.

Textbook sections 12.1–12.3 pp.298–300

Angles, triangles and drawing interpretation

Day 20 • 2026-12-02 • 15 questions

Learner PDF page 6 • W06-L05-Q21

15 An apprentice obtains 179 degrees when adding three rounded angle readings. How far is this total from the exact triangle total?

1 degrees

The exact triangle total is 180 degrees. The difference is 1 degree; inspect the rounding and readings.

Textbook sections 12.1–12.3 pp.298–300

Pythagorean theorem and brace lengths

Day 20 • 2026-12-02 • 15 questions

Learner PDF page 1 • W06-L06-Q01

01 Which side in this Shape Canvas triangle is the hypotenuse?

A. The side opposite the square corner

The hypotenuse is opposite the right angle; its length is not needed to identify it.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Learner PDF page 2 • W06-L06-Q02

02 Use the 3 m and 4 m legs shown. What is the sum of their squares?

25 m²

Square each leg separately, then add: $3^2 + 4^2 = 9 + 16 = 25$ m².

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Learner PDF page 3 • W06-L06-Q03

03 Before checking a brace length with the Numeracy Hub calculator, which expression finds the hypotenuse for 6 m and 8 m legs?

C. sqrt(6² + 8²)

For a hypotenuse, add the squared leg lengths and take the square root of the entire sum.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Learner PDF page 3 • W06-L06-Q04

04 Which condition is required for $a^2 + b^2 = c^2$ in this lesson?

B. The triangle has a right angle

The squared-side relationship applies to right triangles.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Learner PDF page 4 • W06-L06-Q05

05 For this Shape Canvas triangle, find the hypotenuse.

5 m

c = sqrt(3² + 4²) = sqrt(25) = 5 m.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Pythagorean theorem and brace lengths

Day 20 • 2026-12-02 • 15 questions

Learner PDF page 4 • W06-L06-Q09

06 A brace spans perpendicular runs of 5 m and 12 m. Find the brace length.

13 m

The brace is the hypotenuse: sqrt(25 + 144) = sqrt(169) = 13 m.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Learner PDF page 4 • W06-L06-Q10

07 A right triangle has a 10 m hypotenuse and one 6 m leg. Find its other leg.

8 m

Subtract the known leg’s square: sqrt(100 − 36) = sqrt(64) = 8 m.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Learner PDF page 5 • W06-L06-Q11

08 A rectangular panel is 1200 mm by 800 mm as shown. Find its diagonal. Round to 1 decimal place.

1442.2 mm

The diagonal is sqrt(1200^2 + 800^2), approximately 1442.2 mm.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Learner PDF page 5 • W06-L06-Q12

09 A right triangle has legs 900 mm and 1200 mm. Find its diagonal brace.

1500 mm

sqrt(900^2 + 1200^2) = sqrt(2250000) = 1500 mm.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Learner PDF page 5 • W06-L06-Q13

10 A 25 m diagonal and a 7 m leg form a right triangle. Find the other leg.

24 m

sqrt(625 − 49) = sqrt(576) = 24 m.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Pythagorean theorem and brace lengths

Day 20 • 2026-12-02 • 15 questions

Learner PDF page 6 • W06-L06-Q17

11 A learner reports 7 m as the diagonal of a 3 m by 4 m rectangle. What was likely done?

A. Added the two side lengths instead of using squared lengths

3 + 4 = 7 m is the path along two edges. The diagonal is sqrt($3^2 + 4^2$) = 5 m.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Learner PDF page 6 • W06-L06-Q18

12 Check a 13 m hypotenuse with 5 m and 12 m legs. What is $13^2 - 5^2 - 12^2$?

0 m²

169 − 25 − 144 = 0 m², so the squared-side relationship balances.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Learner PDF page 6 • W06-L06-Q19

13 A 6 m and 8 m right-triangle calculation produces 100 m. Which final step was missed?

C. Take the square root

100 is the sum of squared lengths in m². sqrt(100) = 10 m is the length.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Learner PDF page 7 • W06-L06-Q20

14 A proposed hypotenuse is 4.8 m and the longer leg is 5 m. What should you conclude?

B. The result needs checking because the hypotenuse cannot be shorter than a leg

A hypotenuse must exceed each leg. Recheck the formula, values and calculator entry.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Pythagorean theorem and brace lengths

Day 20 • 2026-12-02 • 15 questions

Learner PDF page 7 • W06-L06-Q21

15 A brace spans perpendicular legs of 1.5 m and 2.0 m. Find the brace length in millimetres for the cut list.

2,500 mm

Brace = √(1.5² + 2.0²) = √(2.25 + 4.00) = √6.25 = 2.5 m. In millimetres: 2.5 × 1,000 = 2,500 mm.

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

Checking square, diagonals and layout

Day 21 • 2026-12-03 • 15 questions

Learner PDF page 1 • W06-L07-Q01

01 A 3–4–5 check uses the two legs shown in Shape Canvas. What diagonal makes the corner square?

5 m

With perpendicular legs of 3 m and 4 m, the diagonal is sqrt(25) = 5 m.

Textbook sections 12.5–12.6 pp.304–306

Learner PDF page 2 • W06-L07-Q02

02 Which pair of lines should be measured when checking a rectangular frame?

D. Both corner-to-opposite-corner diagonals

Comparing both diagonals helps detect skew after the side lengths and frame shape are established.

Textbook sections 12.5–12.6 pp.304–306

Learner PDF page 2 • W06-L07-Q03

03 When using a Numeracy Hub measurement diagram, which reading should be recorded?

C. The stated or measured dimension with its unit

Diagram scale on a display is not a physical measuring scale. Record the actual labelled or measured dimension.

Textbook sections 12.5–12.6 pp.304–306

Learner PDF page 3 • W06-L07-Q04

04 The floor in Shape Canvas is 7.2 m by 3.6 m. What diagonal length is expected for a rectangle? Round to 2 decimal places.

8.05 m

Use the side dimensions: sqrt($7.2^2 + 3.6^2$) is approximately 8.05 m.

Textbook sections 12.5–12.6 pp.304–306

Checking square, diagonals and layout

Day 21 • 2026-12-03 • 15 questions

Learner PDF page 3 • W06-L07-Q05

05 Which set is a scaled 3–4–5 check?

A. 900 mm, 1200 mm, 1500 mm

Multiplying 3, 4 and 5 by 300 gives 900, 1200 and 1500 mm.

Textbook sections 12.5–12.6 pp.304–306

Learner PDF page 4 • W06-L07-Q09

06 A scaled 3–4–5 check has a shorter leg of 1.8 m. Find the corresponding longer leg.

2.4 m

The scale factor is 1.8 ÷ 3 = 0.6. The other leg is 4 × 0.6 = 2.4 m.

Textbook sections 12.5–12.6 pp.304–306

Learner PDF page 4 • W06-L07-Q10

07 A 3–4–5 square check has a 1.8 m shorter leg. Find its diagonal target.

3 m

The common factor is 0.6, so the diagonal target is 5 × 0.6 = 3 m.

Textbook sections 12.5–12.6 pp.304–306

Learner PDF page 4 • W06-L07-Q11

08 Two frame diagonals measure 2508 mm and 2496 mm. How much do they differ?

12 mm

Subtract the smaller measurement from the larger: 2508 − 2496 = 12 mm.

Textbook sections 12.5–12.6 pp.304–306

Learner PDF page 5 • W06-L07-Q12

09 A 900 mm square is shown. Find its theoretical diagonal. Round to 1 decimal place.

1272.8 mm

The diagonal of the square is sqrt(900^2 + 900^2), approximately 1272.8 mm.

Textbook sections 12.5–12.6 pp.304–306

Checking square, diagonals and layout

Day 21 • 2026-12-03 • 15 questions

Learner PDF page 5 • W06-L07-Q13

10 A rectangular opening has sides of 1.5 m and 2 m. Find the target diagonal.

2.5 m

sqrt(2.25 + 4) = sqrt(6.25) = 2.5 m.

Textbook sections 12.5–12.6 pp.304–306

Learner PDF page 6 • W06-L07-Q17

11 For a fixed two-leg layout, the distance between the free endpoints exceeds the right-angle target. What does that indicate?

A. The included corner is more open than 90 degrees

With the two leg lengths held fixed, a larger opposite distance indicates a larger included angle.

Textbook sections 12.5–12.6 pp.304–306

Learner PDF page 6 • W06-L07-Q18

12 With two leg lengths held fixed, their endpoint distance is below the right-angle target. How should the corner change?

D. Open the included corner toward 90 degrees

A shorter endpoint distance indicates a corner below 90 degrees; opening it increases that distance toward the target.

Textbook sections 12.5–12.6 pp.304–306

Learner PDF page 6 • W06-L07-Q19

13 A 3 m by 4 m layout shows 5.03 m across the endpoints. How far is this above the square target?

30 mm

The right-angle target is 5 m. The excess is 0.03 m, or 30 mm.

Textbook sections 12.5–12.6 pp.304–306

Checking square, diagonals and layout

Day 21 • 2026-12-03 • 15 questions

Learner PDF page 7 • W06-L07-Q20

14 The two measured diagonals differ. What is a useful next action?

B. Recheck the measurements and adjust the layout before repeating both diagonals

Repeat measurements to rule out reading error, then adjust and check both diagonals again.

Textbook sections 12.5–12.6 pp.304–306

Learner PDF page 7 • W06-L07-Q21

15 A rectangular frame is 2.4 m by 3.2 m. Find the target diagonal for a square check.

4 m

This is a 3–4–5 triangle scaled by 0.8. Diagonal = $\sqrt{(2.4^2 + 3.2^2)} = \sqrt{(5.76 + 10.24)} = \sqrt{16} = 4$ m.

Textbook sections 12.5–12.6 pp.304–306

SOH CAH TOA and missing sides

Day 21 • 2026-12-03 • 15 questions

Learner PDF page 1 • W06-L08-Q01

01 For the angle opposite the 3 m leg in this Shape Canvas triangle, which leg is adjacent?

A. The 4 m leg

The adjacent leg meets the selected angle and is not the hypotenuse. Here it is the 4 m leg.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

Learner PDF page 2 • W06-L08-Q02

02 For the angle opposite the 3 m leg, find opposite divided by adjacent.

0.75

The opposite leg is 3 m and the adjacent leg is 4 m, so their ratio is 3 ÷ 4 = 0.75.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

Learner PDF page 3 • W06-L08-Q03

03 A Numeracy Hub calculator will evaluate sin(30 degrees). Which angle mode should be selected?

C. DEG

The angle is stated in degrees, so the calculation requires degree mode.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

Learner PDF page 3 • W06-L08-Q04

04 Which ratio uses opposite and hypotenuse?

B. Sine

SOH means sine = opposite ÷ hypotenuse.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

Learner PDF page 4 • W06-L08-Q05

05 Which ratio uses adjacent and hypotenuse?

A. Cosine

CAH means cosine = adjacent ÷ hypotenuse.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

SOH CAH TOA and missing sides

Day 21 • 2026-12-03 • 15 questions

Learner PDF page 4 • W06-L08-Q09

06 A right triangle has a 10 m hypotenuse and a 30-degree angle. Find the opposite side. Round to 2 decimal places.

5.00 m

Opposite = hypotenuse × sine: 10 × sin(30°) = 5 m.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

Learner PDF page 4 • W06-L08-Q10

07 A 12 m hypotenuse makes a 60-degree angle with the adjacent leg. Find that adjacent leg. Round to 2 decimal places.

6.00 m

Adjacent = hypotenuse × cosine: 12 × cos(60°) = 6 m.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

Learner PDF page 4 • W06-L08-Q11

08 A right triangle has an adjacent run of 4 m and an angle of 35 degrees. Find the opposite rise. Round to 2 decimal places.

2.80 m

Opposite = adjacent × tan(angle): 4 × tan(35°) is approximately 2.80 m.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

Learner PDF page 5 • W06-L08-Q12

09 The opposite side is 3 m and the angle is 30 degrees. Find the hypotenuse. Round to 2 decimal places.

6.00 m

From sin(angle) = opposite/hypotenuse, hypotenuse = 3 ÷ sin(30°) = 6 m.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

SOH CAH TOA and missing sides

Day 21 • 2026-12-03 • 15 questions

Learner PDF page 5 • W06-L08-Q13

10 The adjacent side is 5 m and the angle is 40 degrees. Find the hypotenuse. Round to 2 decimal places.

6.53 m

Hypotenuse = adjacent ÷ cos(angle): 5 ÷ cos(40°) is approximately 6.53 m.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

Learner PDF page 5 • W06-L08-Q17

11 A learner needs the opposite side and knows the adjacent side and angle. Which equation is suitable?

A. Opposite = adjacent × tan(angle)

Tangent links opposite and adjacent. Rearranging opposite/adjacent = tan(angle) gives the required equation.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

Learner PDF page 6 • W06-L08-Q18

12 A calculated leg exceeds the stated hypotenuse. What is the best response?

D. Recheck the ratio, rearrangement and angle mode

Every leg is shorter than the hypotenuse. A larger result signals an error in setup or entry.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

Learner PDF page 6 • W06-L08-Q19

13 A 10 m hypotenuse at 30 degrees gave an opposite side of 5 m. What ratio checks this result?

0.5

Opposite/hypotenuse = 5/10 = 0.5, matching sin(30°).

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

SOH CAH TOA and missing sides

Day 21 • 2026-12-03 • 15 questions

Learner PDF page 6 • W06-L08-Q20

14 Which is the better first step for an unfamiliar right-triangle question?

B. Mark the angle and label opposite, adjacent and hypotenuse

Correct side labels determine which trigonometric ratio contains the known and unknown sides.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

Learner PDF page 7 • W06-L08-Q21

15 Why should intermediate trig values be kept unrounded?

A. Early rounding can change the final length

Keep calculator precision through the calculation and round the final answer as requested.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

Angles, grades and carpentry slope applications

Day 22 • 2026-12-04 • 15 questions

Learner PDF page 1 • W06-L09-Q01

01 What is the correct definition of percentage grade?

A. Rise ÷ horizontal run × 100

Percentage grade compares vertical rise with horizontal run, using the same units.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Learner PDF page 2 • W06-L09-Q02

02 Using the 3 m rise and 4 m horizontal run in Shape Canvas, calculate the grade.

75 percent

Grade = rise/run × 100 = 3/4 × 100 = 75%.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Learner PDF page 3 • W06-L09-Q03

03 For an angle calculated from a rise/run ratio, which calculator operation is required?

C. Inverse tangent in DEG mode

Angle = arctan(rise/run). Use degree mode when the answer is requested in degrees.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Learner PDF page 3 • W06-L09-Q04

04 Is a 2% grade equal to a 2-degree angle?

B. No; percent grade and degrees are different measures

A 2% grade has rise/run = 0.02, giving an angle of about 1.15 degrees, not 2 degrees.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Learner PDF page 4 • W06-L09-Q05

05 Find the angle above the horizontal for the 3 m rise and 4 m run shown. Round to 2 decimal places.

36.87 degrees

Angle = arctan(3/4), approximately 36.87 degrees.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Angles, grades and carpentry slope applications

Day 22 • 2026-12-04 • 15 questions

Learner PDF page 4 • W06-L09-Q09

06 A rise is 150 mm over a horizontal run of 3000 mm. Find the grade.

5 percent

Grade = 150/3000 × 100 = 5%.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Learner PDF page 4 • W06-L09-Q10

07 A rise is 200 mm over a horizontal run of 4000 mm. Find the angle. Round to 2 decimal places.

2.86 degrees

Use arctan(200/4000); the angle is approximately 2.86 degrees.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Learner PDF page 5 • W06-L09-Q11

08 A classroom slope has a 2% grade and a 6 m horizontal run. Find its rise in millimetres.

120 mm

Rise = grade as a decimal × run = 0.02 × 6 m = 0.12 m = 120 mm.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Learner PDF page 5 • W06-L09-Q12

09 A grade is 8% and the vertical rise is 0.24 m. Find the horizontal run.

3 m

Run = rise/grade ratio = 0.24/0.08 = 3 m.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Learner PDF page 5 • W06-L09-Q13

10 Find the angle corresponding to a 10% grade. Round to 2 decimal places.

5.71 degrees

A 10% grade is a ratio of 0.10. arctan(0.10) is approximately 5.71 degrees.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Angles, grades and carpentry slope applications

Day 22 • 2026-12-04 • 15 questions

Learner PDF page 6 • W06-L09-Q17

11 A learner uses a 5 m sloping length as the run when a horizontal run is given. What should change?

A. Use the horizontal run in the grade denominator

Grade is rise/horizontal run, so substituting sloping length changes the quantity being calculated.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Learner PDF page 6 • W06-L09-Q18

12 A 4% grade over 2500 mm of horizontal run gives what rise?

100 mm

Use 4% = 0.04: rise = 0.04 × 2500 = 100 mm.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Learner PDF page 6 • W06-L09-Q19

13 Which pair correctly labels two different slope results?

C. 5% grade and 2.86-degree angle

A 5% grade gives arctan(0.05), approximately 2.86 degrees. The labels distinguish the quantities.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Learner PDF page 7 • W06-L09-Q20

14 A ramp rises 300 mm over a horizontal run of 3,600 mm. Find the percentage grade. Round to 2 decimal places.

8.33%

Grade = rise ÷ run × 100 = 300 ÷ 3,600 × 100 = 8.3333…, so 8.33%. This is a percentage, not an angle.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Angles, grades and carpentry slope applications

Day 22 • 2026-12-04 • 15 questions

Learner PDF page 7 • W06-L09-Q21

15 An apprentice finds a rise/run ratio of 0.075. What percentage grade should be written?

7.5 percent

Multiply the ratio by 100: 0.075 × 100 = 7.5%.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Carpentry drawing: floor layout and cut list

Day 23 • 2026-12-07 • 15 questions

Learner PDF page 1 • W07-L01-Q01

01 What is the perimeter of the 7.2 m by 3.6 m floor shown?

21.6 m

Perimeter = 2(length + width) = 2(7.2 + 3.6) = 21.6 m.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Learner PDF page 2 • W07-L01-Q02

02 What is the area of the 7.2 m by 3.6 m floor shown?

25.92 m²

Area = length × width = 7.2 × 3.6 = 25.92 m².

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Learner PDF page 3 • W07-L01-Q03

03 The Numeracy Hub area/perimeter diagram is open. Which quantity is needed for flooring coverage?

C. Area

Coverage is a surface quantity and uses square units, so it requires area.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Learner PDF page 3 • W07-L01-Q04

04 A separate stock sketch shows a length of 4500 mm. Should this be combined directly with a floor area of 25.92 m²?

B. No; they are different quantities and contexts

A linear stock quantity and a floor area are not directly compatible. Keep their equations and units separate.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Learner PDF page 4 • W07-L01-Q05

05 Convert the 7.2 m floor length shown into millimetres.

7200 mm

There are 1000 mm per metre: 7.2 × 1000 = 7200 mm.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Carpentry drawing: floor layout and cut list

Day 23 • 2026-12-07 • 15 questions

Learner PDF page 4 • W07-L01-Q09

06 Four pieces are each 900 mm long. Find their combined finished length before kerf.

3600 mm

Four equal pieces require 4 × 900 = 3600 mm of finished material.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Learner PDF page 4 • W07-L01-Q10

07 Four cuts each remove 3 mm of kerf. What total length is lost to kerf?

12 mm

Multiply the stated number of cuts by kerf per cut: 4 × 3 = 12 mm.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Learner PDF page 5 • W07-L01-Q11

08 From 4500 mm stock, cut four 900 mm pieces using four cuts of 3 mm kerf. What offcut remains?

888 mm

Offcut = stock − finished pieces − stated kerfs = 4500 − 3600 − 12 = 888 mm.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Learner PDF page 5 • W07-L01-Q12

09 A cut list needs two 1250 mm pieces and three 600 mm pieces. Find the total finished length.

4300 mm

2 × 1250 + 3 × 600 = 2500 + 1800 = 4300 mm.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Learner PDF page 5 • W07-L01-Q13

10 Find the diagonal of the 7.2 m by 3.6 m floor. Round to 2 decimal places.

8.05 m

Diagonal = sqrt($7.2^2 + 3.6^2$), approximately 8.05 m.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Carpentry drawing: floor layout and cut list

Day 23 • 2026-12-07 • 15 questions

Learner PDF page 6 • W07-L01-Q17

11 A learner subtracts four pieces from stock but omits four stated kerfs. What happens to the reported offcut?

A. It is too large by the total kerf

Kerf consumes material. Omitting it leaves too much stock in the arithmetic.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Learner PDF page 6 • W07-L01-Q18

12 Why must the number of cuts be stated rather than guessed from the number of pieces?

D. The cutting sequence and usable end can change the count

Use the problem’s specified cut count or document the cutting sequence before applying kerf.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Learner PDF page 6 • W07-L01-Q19

13 Four 3 mm kerfs were omitted from an offcut calculation. By how many millimetres must the reported offcut be reduced?

12 mm

The omitted loss is 4 × 3 = 12 mm; subtract that from the original offcut result.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Learner PDF page 7 • W07-L01-Q20

14 Which check best catches a cut-list addition error?

B. Add finished lengths, total kerf and offcut back to the original stock length

The material balance should recover the original stock length when all consumed and remaining lengths are included.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Carpentry drawing: floor layout and cut list

Day 23 • 2026-12-07 • 15 questions

Learner PDF page 7 • W07-L01-Q21

15 A stock balance has 3600 mm of finished pieces, 12 mm kerf and 888 mm offcut. What original stock length does this verify?

4500 mm

Add finished pieces, kerf and offcut: 3600 + 12 + 888 = 4500 mm of original stock.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Materials practice: sheeting and allowance

Day 23 • 2026-12-07 • 15 questions

Learner PDF page 1 • W07-L02-Q01

01 Find the net area of the 7.2 m by 3.6 m floor shown.

25.92 m²

The floor area is 7.2 × 3.6 = 25.92 m².

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Learner PDF page 1 • W07-L02-Q02

02 A rectangular sheet is 2.4 m by 1.2 m. What area does one sheet cover before waste?

2.88 m²

Sheet area = 2.4 × 1.2 = 2.88 m².

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Learner PDF page 2 • W07-L02-Q03

03 A Numeracy Hub area diagram is used to estimate coating. What must be known in addition to area?

C. The stated coverage per tin or litre

Area divided by the stated coverage determines the required coating quantity.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Learner PDF page 3 • W07-L02-Q04

04 Using 2.88 m² per sheet, how many sheets equal the shown floor’s 25.92 m² net area?

9 sheets

25.92 ÷ 2.88 = 9 sheets before any allowance; the separate layout must also be checked.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Materials practice: sheeting and allowance

Day 23 • 2026-12-07 • 15 questions

Learner PDF page 3 • W07-L02-Q05

05 A class exercise states a 10% allowance. What quantity is multiplied by 0.10?

A. The stated net requirement

The allowance is 10% of the stated net requirement; add it before the final purchase rounding.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Learner PDF page 4 • W07-L02-Q09

06 Apply a stated 10% classroom allowance to a 9-sheet base. What is the unrounded requirement?

9.9 sheets

Add 10% by multiplying by 1.10: 9 × 1.10 = 9.9 sheets.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Learner PDF page 4 • W07-L02-Q10

07 The unrounded requirement is 9.9 sheets and only whole sheets are sold. How many should be ordered?

10 sheets

Round up to ensure enough material: 9.9 requires 10 whole sheets.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Learner PDF page 4 • W07-L02-Q11

08 A wall has net area 42 m². Add the stated 8% classroom allowance. What total area should the estimate use?

45.36 m²

42 × 1.08 = 45.36 m², including the stated allowance.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Materials practice: sheeting and allowance

Day 23 • 2026-12-07 • 15 questions

Learner PDF page 4 • W07-L02-Q12

09 One tin covers 12 m². A surface needs 53 m² of coverage in one coat. How many whole tins are required?

5 tins

53 ÷ 12 is about 4.42 tins. Round up to 5 whole tins.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Learner PDF page 5 • W07-L02-Q13

10 A 30 m² surface needs two coats. Each tin covers 18 m² for one coat. Find the whole-tin quantity.

4 tins

Total coverage is 30 × 2 = 60 m². 60/18 is about 3.33, so buy 4 tins.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Learner PDF page 5 • W07-L02-Q17

11 Why should 9.9 sheets be rounded up rather than to the nearest whole number as a general purchase rule?

A. A fractional sheet requirement still needs a full purchased sheet

The purchasing unit is a whole sheet, so a positive fraction means another sheet is required.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Learner PDF page 5 • W07-L02-Q18

12 A floor needs 20 sheets before allowance. Apply a stated 10% classroom allowance. How many extra sheets is that?

2 sheets

10% of 20 = 0.10 × 20 = 2 extra sheets. The order is 20 + 2 = 22 sheets.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Materials practice: sheeting and allowance

Day 23 • 2026-12-07 • 15 questions

Learner PDF page 6 • W07-L02-Q19

13 A base of 9 sheets becomes an order of 10 sheets. How many whole sheets above the base is that?

1 sheets

The rounded order exceeds the no-waste base by 10 − 9 = 1 sheet.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Learner PDF page 6 • W07-L02-Q20

14 A learner rounds each partial calculation upward several times. What is the better approach for a material estimate?

B. Keep intermediate precision and round the final purchase quantity upward

Repeated early rounding can unnecessarily inflate an estimate. Apply the stated allowance, then round the final purchase unit.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Learner PDF page 6 • W07-L02-Q21

15 The 9-sheet base covers 25.92 m². What area does a stated 10% allowance add?

2.592 m²

The allowance area is 25.92 × 0.10 = 2.592 m².

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Skill clinics: practise your priority topic

Day 24 • 2026-12-10 • 15 questions

Learner PDF page 1 • W07-L03-Q01

01 A learner writes 1/2 + 1/4 = 2/6. What correction addresses the error?

A. Use a common denominator before adding

Rewrite 1/2 as 2/4, then add 2/4 + 1/4 = 3/4.

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Learner PDF page 1 • W07-L03-Q02

02 A learner added 1/2 in + 1/4 in and wrote 2/6 in. Find the correct sum as a fraction in inches.

3/4 in

The learner added the denominators. Rename 1/2 as 2/4. Then 2/4 + 1/4 = 3/4 in.

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Learner PDF page 1 • W07-L03-Q03

03 Correct the unit conversion: 2.4 m equals how many millimetres?

2400 mm

Multiply metres by 1000: 2.4 × 1000 = 2400 mm.

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Learner PDF page 2 • W07-L03-Q04

04 For the 1200 mm by 800 mm panel shown, which expression finds perimeter?

B. 2 × (1200 + 800)

Perimeter adds all four sides; for a rectangle this is 2(length + width).

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Learner PDF page 2 • W07-L03-Q05

05 A floor is 4.2 m by 3.5 m. A learner wrote the area as 14.7 m³. What is the correct area with the correct unit?

B. 14.7 m²

Area = 4.2 × 3.5 = 14.7. Area uses square units, so the answer is 14.7 m², not m³.

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Skill clinics: practise your priority topic

Day 24 • 2026-12-10 • 15 questions

Learner PDF page 3 • W07-L03-Q09

06 Retry with new numbers: calculate 3/4 + 1/8.

7/8

Rewrite 3/4 as 6/8. Then 6/8 + 1/8 = 7/8.

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Learner PDF page 3 • W07-L03-Q10

07 Retry a regrouping problem: calculate 2 1/4 − 3/4.

1 1/2

Rewrite 2 1/4 as 1 5/4, then subtract 3/4 to get 1 1/2.

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Learner PDF page 3 • W07-L03-Q11

08 A rectangle has area 18 m² and length 6 m. Find its width.

3 m

From A = L × W, divide by length: W = 18/6 = 3 m.

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Learner PDF page 4 • W07-L03-Q12

09 Find the area of the 7.2 m by 3.6 m floor shown.

25.92 m²

Use A = length × width: 7.2 × 3.6 = 25.92 m².

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Learner PDF page 4 • W07-L03-Q13

10 Convert 75 mm to metres.

0.075 m

Divide millimetres by 1000: 75/1000 = 0.075 m.

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Skill clinics: practise your priority topic

Day 24 • 2026-12-10 • 15 questions

Learner PDF page 4 • W07-L03-Q17

11 Fresh check: add 2 3/8 in and 1 1/2 in.

3 7/8 in

Rename 1 1/2 as 1 4/8. Then 2 3/8 + 1 4/8 = 3 7/8 in.

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Learner PDF page 5 • W07-L03-Q18

12 Convert 1,750 mm to metres.

1.75 m

1 m = 1,000 mm. Divide: 1,750 ÷ 1,000 = 1.75 m.

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Learner PDF page 5 • W07-L03-Q19

13 A corrected width is 3 m and the length is 6 m. What area checks the result?

18 m²

Substitute into A = L × W: 6 × 3 = 18 m².

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Learner PDF page 5 • W07-L03-Q20

14 Fresh check: a right triangle has legs of 9 m and 12 m. Find the hypotenuse.

15 m

√(9² + 12²) = √(81 + 144) = √225 = 15 m.

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Learner PDF page 5 • W07-L03-Q21

15 Convert 6 in to millimetres using exactly 25.4 mm per inch.

152.4 mm

Inches to millimetres: multiply. 6 × 25.4 = 152.4 mm.

Textbook Chapter 4–5 pp.88–132; Chapter 7–8 pp.164–204; Chapter 12–13 pp.298–333, selected

Mixed-topic calculation practice

Day 24 • 2026-12-10 • 15 questions

Learner PDF page 1 • W07-L04-Q01

01 What should you identify before calculating a mixed-topic question?

A. The required quantity, unit and relevant given values

Reading the required result prevents choosing an operation merely because certain numbers are visible.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Learner PDF page 2 • W07-L04-Q02

02 A Numeracy Hub question requests a length to two decimal places. When should the final rounding occur?

D. After completing the calculation with full intermediate precision

Use full intermediate precision and apply the stated rounding to the final result.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Learner PDF page 3 • W07-L04-Q03

03 The 1200 mm by 800 mm panel needs edge trim. Which quantity is required?

C. Perimeter

Edge trim follows the outside boundary, so calculate perimeter.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Learner PDF page 4 • W07-L04-Q04

04 The 7.2 m by 3.6 m floor needs surface covering. Which quantity is required?

B. Area

Surface covering is based on area and uses square units.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Learner PDF page 4 • W07-L04-Q05

05 A slab form is 2 m long, 1.5 m wide and 0.2 m deep. What is its volume?

C. 0.6 m³

$V = L \times W \times H = 2 \times 1.5 \times 0.2 = 0.6$ m³.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Mixed-topic calculation practice

Day 24 • 2026-12-10 • 15 questions

Learner PDF page 5 • W07-L04-Q09

06 Calculate 7 + 3 × 6.

25

Multiply first: 3 × 6 = 18. Then 7 + 18 = 25.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Learner PDF page 5 • W07-L04-Q10

07 Calculate 5/8 + 1/4.

7/8

Use eighths: 5/8 + 2/8 = 7/8.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Learner PDF page 5 • W07-L04-Q11

08 Convert 3.5 inches to millimetres using 25.4 mm per inch.

88.9 mm

3.5 × 25.4 = 88.9 mm.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Learner PDF page 5 • W07-L04-Q12

09 A rectangle’s area is 30 m² and its length is 7.5 m. Find its width.

4 m

Width = area/length = 30/7.5 = 4 m.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Learner PDF page 6 • W07-L04-Q13

10 A right triangle has legs 7 m and 24 m. Find its hypotenuse.

25 m

sqrt(49 + 576) = sqrt(625) = 25 m.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Mixed-topic calculation practice

Day 24 • 2026-12-10 • 15 questions

Learner PDF page 6 • W07-L04-Q17

11 A solution has a correct number but no units. What should be added?

A. The unit for the requested quantity

The unit identifies what the number measures and is part of a complete answer.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Learner PDF page 6 • W07-L04-Q18

12 Calculate 5 + 3 × 4.

17

Multiply first: 3 × 4 = 12. Then 5 + 12 = 17.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Learner PDF page 6 • W07-L04-Q19

13 A result claims that 2.5 m equals 250 mm. What is the correct millimetre value?

2500 mm

Metres convert to millimetres by multiplying by 1000: 2.5 × 1000 = 2500 mm.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Learner PDF page 7 • W07-L04-Q20

14 Fresh check: find 15% of $240.

$36

15% = 0.15. 0.15 × 240 = $36.

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Learner PDF page 7 • W07-L04-Q21

15 Check a 4 m width for a 7.5 m-long rectangle by calculating its area.

30 m²

Substitute into A = L × W: 7.5 × 4 = 30 m².

Textbook formula sheet p.361; conversion factors pp.362–363; relevant chapters as needed

Review: numbers, units and formulas

Day 25 • 2026-12-14 • 15 questions

Learner PDF page 1 • W08-L01-Q01

01 Calculate the total length of six 725 mm pieces.

4350 mm

Six equal pieces require 6 × 725 = 4350 mm.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Learner PDF page 1 • W08-L01-Q02

02 Calculate 1 3/4 + 2 1/8.

3 7/8

Use eighths: 1 6/8 + 2 1/8 = 3 7/8.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Learner PDF page 2 • W08-L01-Q03

03 Numeracy Hub is used to check 24 − 3 × 5. Which paper method is correct?

C. Multiply 3 × 5 before subtracting from 24

Order of operations gives 3 × 5 = 15, then 24 − 15 = 9.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Learner PDF page 2 • W08-L01-Q04

04 Calculate 5.75 + 2.6 − 1.85.

6.5

Align decimal places: 5.75 + 2.60 = 8.35; 8.35 − 1.85 = 6.50.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Learner PDF page 3 • W08-L01-Q05

05 A net length is 64 m. Add a stated 12.5% allowance to find the total.

72 m

12.5% is one eighth, so the allowance is 8 m. The total is 64 + 8 = 72 m.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Review: numbers, units and formulas

Day 25 • 2026-12-14 • 15 questions

Learner PDF page 3 • W08-L01-Q09

06 Convert the 1200 mm length in the Shape Canvas panel to metres.

1.2 m

Divide by 1000: 1200 mm = 1.2 m.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Learner PDF page 4 • W08-L01-Q10

07 Convert the 18.5-inch length shown into millimetres using 25.4 mm per inch.

469.9 mm

18.5 × 25.4 = 469.9 mm.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Learner PDF page 4 • W08-L01-Q11

08 A 1:20 drawing shows a length of 65 mm. Find the full-size length in millimetres.

1300 mm

Full-size length = drawing length × scale factor = 65 × 20 = 1300 mm.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Learner PDF page 4 • W08-L01-Q12

09 A rectangle has area 32 m² and width 4 m. Find its length.

8 m

From A = L × W, L = A/W = 32/4 = 8 m.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Learner PDF page 5 • W08-L01-Q13

10 Use A = L × W for a rectangle with L = 4.5 m and W = 2 m. Find A.

9 m²

Substitute into the formula: A = 4.5 × 2 = 9 m².

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Review: numbers, units and formulas

Day 25 • 2026-12-14 • 15 questions

Learner PDF page 5 • W08-L01-Q17

11 A learner gets 120 mm when converting 1.2 m. Which check catches the error?

A. One metre is already 1000 mm, so 1.2 m must exceed 1000 mm

A magnitude check catches the wrong conversion factor before detailed recalculation.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Learner PDF page 5 • W08-L01-Q18

12 Which equation checks a width found from area and length?

D. Length × calculated width = original area

Substitute the calculated width into the original A = L × W relationship.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Learner PDF page 6 • W08-L01-Q19

13 A calculated width is 5 m and length is 8 m. What perimeter checks the result?

26 m

Substitute into P = 2(L + W): 2(8 + 5) = 26 m.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Learner PDF page 6 • W08-L01-Q20

14 What is a good estimate for 49 × 21 before using a calculator?

B. About 1000

Rounding to 50 × 20 gives 1000, which checks the magnitude of the exact answer.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Learner PDF page 6 • W08-L01-Q21

15 The exact product 49 × 21 is how much greater than the estimate 1000?

29

49 × 21 = 1029. The difference from 1000 is 29.

Textbook Chapter 1–8, selected; formula sheet p.361; conversion factors pp.362–363

Review: geometry and job problems

Day 25 • 2026-12-14 • 15 questions

Learner PDF page 1 • W08-L02-Q01

01 The shown floor needs edge trim. Which formula applies?

A. P = 2(L + W)

Edge trim follows the rectangle’s boundary, so use its perimeter.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Learner PDF page 2 • W08-L02-Q02

02 The shown rectangular prism is being filled. Which quantity is needed?

D. Volume

A three-dimensional filled space uses volume.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Learner PDF page 3 • W08-L02-Q03

03 Numeracy Hub displays an area/perimeter diagram. Which unit describes surface coverage?

C. m²

Area measures surface coverage and uses square units.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Learner PDF page 3 • W08-L02-Q04

04 A brace crosses a rectangular panel diagonally. Which method finds its length from the two side dimensions?

B. Pythagorean theorem

The diagonal and the perpendicular rectangle sides form a right triangle.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Learner PDF page 4 • W08-L02-Q05

05 For a cylinder, which two measurements determine $V = \pi r^2 h$?

A. Radius and height

The formula uses the radius of the circular base and the perpendicular cylinder height.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Review: geometry and job problems

Day 25 • 2026-12-14 • 15 questions

Learner PDF page 4 • W08-L02-Q09

06 Calculate the perimeter of the 7.2 m by 3.6 m floor shown.

21.6 m

2(7.2 + 3.6) = 21.6 m.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Learner PDF page 4 • W08-L02-Q10

07 A wall is 5 m by 2.4 m with a 1.8 m² opening. Find its net area.

10.2 m²

Gross area is 12 m². Subtract the opening: 12 − 1.8 = 10.2 m².

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Learner PDF page 5 • W08-L02-Q11

08 Find the volume of the shown prism: 2.4 m by 1.2 m by 0.6 m.

1.728 m³

V = L × W × H = 2.4 × 1.2 × 0.6 = 1.728 m³.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Learner PDF page 5 • W08-L02-Q12

09 A right triangle has legs of 8 m and 15 m. Find its hypotenuse.

17 m

sqrt(64 + 225) = sqrt(289) = 17 m.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Learner PDF page 6 • W08-L02-Q13

10 The cylinder shown has radius 0.5 m and height 1.2 m. Use pi = 3.14 to find its volume.

0.942 m³

V = πr²h = 3.14 × 0.5² × 1.2 = 0.942 m³.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Review: geometry and job problems

Day 25 • 2026-12-14 • 15 questions

Learner PDF page 6 • W08-L02-Q17

11 A box-volume calculation is labelled m². What must change?

A. Use m³ for the volume unit

Volume multiplies three lengths and therefore uses cubic units.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Learner PDF page 7 • W08-L02-Q18

12 A 1.728 m³ volume must be expressed in litres. Use 1000 L per m³.

1728 L

1.728 × 1000 = 1728 L.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Learner PDF page 7 • W08-L02-Q19

13 The radius is 0.5 m but a learner substitutes 1 m into πr²h. Which mistake occurred?

C. The diameter was used as the radius

A radius is half the diameter. Doubling r before squaring makes the base area four times too large.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Learner PDF page 7 • W08-L02-Q20

14 From 3000 mm stock, cut two 1200 mm pieces using two 3 mm kerfs. What offcut remains?

594 mm

Subtract finished lengths and stated kerf: 3000 − 2400 − 6 = 594 mm.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Learner PDF page 8 • W08-L02-Q21

15 A diagonal is shorter than the rectangle’s longest side. What does this show?

A. The calculation or measurement needs checking

The diagonal is the hypotenuse of a right triangle and must exceed each side.

Textbook Chapter 9–13 pp.220–339, selected; formula sheet p.361

Readiness check: 12 core questions

Day 26 • 2026-12-16 • 12 questions

Learner PDF page 1 • W08-L03-Q01

01 Calculate 2 3/8 + 1 1/4. Record a simplified fraction or mixed number.

3 5/8

Use eighths: 2 3/8 + 1 2/8 = 3 5/8.

Textbook formula sheet p.361; conversion factors pp.362–363

Learner PDF page 1 • W08-L03-Q02

02 Which decimal is exactly equal to 5/8?

D. 0.625

Divide numerator by denominator: 5 ÷ 8 = 0.625.

Textbook formula sheet p.361; conversion factors pp.362–363

Learner PDF page 2 • W08-L03-Q03

03 A net quantity is 40 m. Apply a stated 10% allowance and report the total.

44 m

Allowance = 40 × 0.10 = 4 m. Total = 44 m.

Textbook formula sheet p.361; conversion factors pp.362–363

Learner PDF page 2 • W08-L03-Q04

04 A 1:25 drawing shows 72 mm. Find the full-size length in millimetres.

1800 mm

Multiply drawing length by the scale factor: 72 × 25 = 1800 mm.

Textbook formula sheet p.361; conversion factors pp.362–363

Learner PDF page 2 • W08-L03-Q09

05 Convert 5 inches to millimetres using 25.4 mm per inch.

127 mm

5 × 25.4 = 127 mm.

Textbook formula sheet p.361; conversion factors pp.362–363

Readiness check: 12 core questions

Day 26 • 2026-12-16 • 12 questions

Learner PDF page 2 • W08-L03-Q10

06 A rectangle has area 27 m² and length 9 m. Find the width.

3 m

Rearrange A = L × W to W = A/L = 27/9 = 3 m.

Textbook formula sheet p.361; conversion factors pp.362–363

Learner PDF page 3 • W08-L03-Q11

07 Find the perimeter of the 1200 mm by 800 mm panel shown.

4000 mm

2(1200 + 800) = 4000 mm.

Textbook formula sheet p.361; conversion factors pp.362–363

Learner PDF page 4 • W08-L03-Q12

08 The shown floor is 7.2 m by 3.6 m. Subtract a 1.2 m by 0.9 m opening to find the net area.

24.84 m²

Gross area = 25.92 m²; opening = 1.08 m²; net = 24.84 m².

Textbook formula sheet p.361; conversion factors pp.362–363

Learner PDF page 5 • W08-L03-Q17

09 Calculate the volume of the shown 2.4 m by 1.2 m by 0.6 m prism.

1.728 m³

V = L × W × H = 2.4 × 1.2 × 0.6 = 1.728 m³.

Textbook formula sheet p.361; conversion factors pp.362–363

Learner PDF page 5 • W08-L03-Q18

10 A right triangle has perpendicular legs 9 m and 12 m. Find its hypotenuse.

15 m

sqrt(81 + 144) = sqrt(225) = 15 m.

Textbook formula sheet p.361; conversion factors pp.362–363

Readiness check: 12 core questions

Day 26 • 2026-12-16 • 12 questions

Learner PDF page 5 • W08-L03-Q19

11 A right triangle has a 14 m hypotenuse and a 30-degree angle. Find the side opposite that angle. Round to 2 decimal places.

7.00 m

Opposite = hypotenuse × sin(angle) = 14 × sin(30°) = 7 m.

Textbook formula sheet p.361; conversion factors pp.362–363

Learner PDF page 6 • W08-L03-Q20

12 Find the angle above horizontal for a rise of 300 mm and horizontal run of 1200 mm. Round to 2 decimal places.

14.04 degrees

Angle = arctan(rise/run) = arctan(0.25), approximately 14.04 degrees.

Textbook formula sheet p.361; conversion factors pp.362–363

Corrections and a personal practice plan

Day 26 • 2026-12-16 • 15 questions

Learner PDF page 1 • W08-L04-Q01

01 A learner got 1/3 + 1/6 = 2/9. Which correction explains the error?

A. Rewrite 1/3 as 2/6 before adding

The denominators must match: 2/6 + 1/6 = 3/6 = 1/2.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Learner PDF page 1 • W08-L04-Q02

02 A learner converted 3 in to millimetres as 3 ÷ 25.4. Find the correct value.

76.2 mm

Inches to millimetres multiplies by 25.4: 3 × 25.4 = 76.2 mm. Dividing gave a tiny number, which fails an estimate check.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Learner PDF page 2 • W08-L04-Q03

03 For the shown panel, a learner wrote area = 2(1200 + 800). What was actually calculated?

C. Perimeter

2(length + width) is the rectangle’s perimeter formula. Area requires length × width.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Learner PDF page 2 • W08-L04-Q04

04 A learner used $10^2 + 6^2$ to find a leg when 10 m is the hypotenuse. Which correction is needed?

B. Use sqrt($10^2 - 6^2$)

For a missing leg, subtract the known leg’s square from the hypotenuse’s square, then take the root.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Corrections and a personal practice plan

Day 26 • 2026-12-16 • 15 questions

Learner PDF page 3 • W08-L04-Q05

05 A learner used a pier’s 0.6 m diameter as its radius. The pier is 1.2 m deep. Find the correct volume with π = 3.14. Round to 3 decimal places.

0.339 m³

Radius = 0.6 ÷ 2 = 0.3 m. V = πr²h = 3.14 × 0.3² × 1.2 = 0.33912 m³, so 0.339 m³.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Learner PDF page 3 • W08-L04-Q09

06 Fresh fraction check: calculate 3 1/4 − 1 5/8.

1 5/8

Use eighths and regroup: 3 2/8 − 1 5/8 = 1 5/8.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Learner PDF page 3 • W08-L04-Q10

07 Fresh conversion check: convert 6.5 inches to millimetres using 25.4 mm per inch.

165.1 mm

6.5 × 25.4 = 165.1 mm.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Learner PDF page 4 • W08-L04-Q11

08 Fresh area check: find the area of the 900 mm square shown.

810000 mm²

Area of a square is side squared: 900 × 900 = 810000 mm².

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Learner PDF page 4 • W08-L04-Q12

09 Fresh missing-leg check: a right triangle has hypotenuse 13 m and one leg 5 m. Find the other leg.

12 m

sqrt(169 − 25) = sqrt(144) = 12 m.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Corrections and a personal practice plan

Day 26 • 2026-12-16 • 15 questions

Learner PDF page 4 • W08-L04-Q13

10 Fresh formula check: area is 42 m² and length is 7 m. Find the width.

6 m

W = A/L = 42/7 = 6 m.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Learner PDF page 5 • W08-L04-Q17

11 Fresh check: find the area of a 3.5 m by 2 m panel.

7 m²

A = L × W = 3.5 × 2 = 7 m².

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Learner PDF page 5 • W08-L04-Q18

12 Fresh check: convert 2.4 m to millimetres.

2,400 mm

1 m = 1,000 mm. 2.4 × 1,000 = 2,400 mm.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Learner PDF page 5 • W08-L04-Q19

13 A corrected width is 6 m and length is 7 m. Calculate the area to verify the width.

42 m²

Substitute into A = L × W: 6 × 7 = 42 m².

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Learner PDF page 5 • W08-L04-Q20

14 Fresh check: find the perimeter of a room 4.2 m long and 2.8 m wide.

14 m

P = 2(L + W) = 2(4.2 + 2.8) = 2 × 7 = 14 m.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Corrections and a personal practice plan

Day 26 • 2026-12-16 • 15 questions

Learner PDF page 6 • W08-L04-Q21

15 Fresh check: a box is 1.2 m by 0.8 m by 0.5 m. Find its volume.

0.48 m³

V = 1.2 × 0.8 × 0.5 = 0.48 m³.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368